Slip 2
Q1) Create an application that allows the user to enter a number in the textbox. Check whether the number in the textbox is perfect number or not. Print the message using Toast control.

Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <TextView
        android:id="@+id/textView"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginStart="16dp"
        android:layout_marginTop="16dp"
        android:layout_marginEnd="16dp"
        android:layout_marginBottom="150dp"
        android:text="Check Perfect Number OR Not"
        android:textSize="20sp"
        app:layout_constraintBottom_toTopOf="@+id/buttonCheck"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintVertical_bias="0.096" />

    <EditText
        android:id="@+id/editTextNumber"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginTop="84dp"
        android:layout_marginBottom="113dp"
        android:hint="Enter a number"
        android:inputType="number"
        app:layout_constraintBottom_toTopOf="@+id/buttonCheck"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toBottomOf="@+id/textView"
        tools:ignore="MissingConstraints" />

    <Button
        android:id="@+id/buttonCheck"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_below="@id/editTextNumber"
        android:layout_centerHorizontal="true"
        android:layout_marginStart="161dp"
        android:layout_marginTop="50dp"
        android:layout_marginEnd="162dp"
        android:layout_marginBottom="300dp"
        android:text="Check"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toBottomOf="@+id/editTextNumber"
        tools:ignore="MissingConstraints" />

</androidx.constraintlayout.widget.ConstraintLayout>
```

MainActivity.java

```java
package com.example.perfectnumber;

import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.Toast;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        EditText editTextNumber = findViewById(R.id.editTextNumber);
        Button buttonCheck = findViewById(R.id.buttonCheck);
        buttonCheck.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v)
            {
                String input = editTextNumber.getText().toString();
                if (!input.isEmpty())
                {
                    int number = Integer.parseInt(input);
                    if (isPerfectNumber(number)) {
                        showToast("The number is a perfect number.");
                    }
                    else
                    {
                        showToast("The number is not a perfect number.");
                    }
                } else
                {
                    showToast("Please enter a number.");
                }
            }
        });
    }
    private boolean isPerfectNumber(int number)
    {
        int sum=0;
        for(int i=1; i<=number/2; i++)
        {
            if(number%i==0)
            {
                sum+=i;
            }
        }
        return sum==number;
    }
    private void showToast(String message)
    {
        Toast.makeText(this, message, Toast.LENGTH_SHORT).show();
    }
}
```

Q.2) Java Android Program to perform all arithmetic Operations using Calculator.

activity_main.xml

```xml
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
```

```
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <!-- Display TextView for the Calculator -->
    <TextView
        android:id="@+id/display"
        android:layout_width="0dp"
        android:layout_height="wrap_content"
        android:text=""
        android:textSize="36sp"
        android:gravity="end"
        android:padding="20dp"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintBottom_toTopOf="@+id/gridLayout"/>

    <!-- Grid Layout for calculator buttons -->
    <GridLayout
        android:id="@+id/gridLayout"
        android:layout_width="0dp"
        android:layout_height="0dp"
        android:columnCount="4"
        android:rowCount="5"
        app:layout_constraintTop_toBottomOf="@id/display"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintBottom_toBottomOf="parent">

        <!-- First Row -->
        <Button android:id="@+id/button1" android:text="1"/>
        <Button android:id="@+id/button2" android:text="2"/>
        <Button android:id="@+id/button3" android:text="3"/>
        <Button android:id="@+id/buttonDivide" android:text="/"/>

        <!-- Second Row -->
        <Button android:id="@+id/button4" android:text="4"/>
        <Button android:id="@+id/button5" android:text="5"/>
        <Button android:id="@+id/button6" android:text="6"/>
        <Button android:id="@+id/buttonMultiply" android:text="*"/>

        <!-- Third Row -->
        <Button android:id="@+id/button7" android:text="7"/>
        <Button android:id="@+id/button8" android:text="8"/>
        <Button android:id="@+id/button9" android:text="9"/>
        <Button android:id="@+id/buttonSubtract" android:text="-"/>

        <!-- Fourth Row -->
        <Button android:id="@+id/button0" android:text="0"/>
        <Button android:id="@+id/buttonClear" android:text="C"/>
        <Button android:id="@+id/buttonEqual" android:text="="/>
        <Button android:id="@+id/buttonAdd" android:text="+"/>
    </GridLayout>
</androidx.constraintlayout.widget.ConstraintLayout>

## MainActivity.java
package com.example.calci;

import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.TextView;
```

```java
import android.widget.Toast;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    private TextView display;
    private String input = "";
    private String operator = "";
    private double firstOperand = 0;
    private double secondOperand = 0;
    private boolean operatorClicked = false;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        display = findViewById(R.id.display);


        Button button0 = findViewById(R.id.button0);
        Button button1 = findViewById(R.id.button1);
        Button button2 = findViewById(R.id.button2);
        Button button3 = findViewById(R.id.button3);
        Button button4 = findViewById(R.id.button4);
        Button button5 = findViewById(R.id.button5);
        Button button6 = findViewById(R.id.button6);
        Button button7 = findViewById(R.id.button7);
        Button button8 = findViewById(R.id.button8);
        Button button9 = findViewById(R.id.button9);

        Button buttonAdd = findViewById(R.id.buttonAdd);
        Button buttonSubtract = findViewById(R.id.buttonSubtract);
        Button buttonMultiply = findViewById(R.id.buttonMultiply);
        Button buttonDivide = findViewById(R.id.buttonDivide);
        Button buttonClear = findViewById(R.id.buttonClear);
        Button buttonEqual = findViewById(R.id.buttonEqual);


        button0.setOnClickListener(v -> appendToInput("0"));
        button1.setOnClickListener(v -> appendToInput("1"));
        button2.setOnClickListener(v -> appendToInput("2"));
        button3.setOnClickListener(v -> appendToInput("3"));
        button4.setOnClickListener(v -> appendToInput("4"));
        button5.setOnClickListener(v -> appendToInput("5"));
        button6.setOnClickListener(v -> appendToInput("6"));
        button7.setOnClickListener(v -> appendToInput("7"));
        button8.setOnClickListener(v -> appendToInput("8"));
        button9.setOnClickListener(v -> appendToInput("9"));
        buttonAdd.setOnClickListener(v -> setOperator("+"));
        buttonSubtract.setOnClickListener(v -> setOperator("-"));
        buttonMultiply.setOnClickListener(v -> setOperator("*"));
        buttonDivide.setOnClickListener(v -> setOperator("/"));
        buttonClear.setOnClickListener(v -> clearInput());

        buttonEqual.setOnClickListener(v -> calculateResult());
    }

    private void appendToInput(String value) {
        if (operatorClicked) {
            input = "";
            operatorClicked = false;
        }
        input += value;
        display.setText(input);
```

```
    }

    private void setOperator(String operator) {
        if (!input.isEmpty()) {
            firstOperand = Double.parseDouble(input);
            this.operator = operator;
            input = "";
            operatorClicked = true;
        }
    }

    private void clearInput() {
        input = "";
        display.setText(input);
        firstOperand = 0;
        secondOperand = 0;
        operator = "";
    }

    private void calculateResult() {
        if (!input.isEmpty()) {
            secondOperand = Double.parseDouble(input);
            double result = 0;

            switch (operator) {
                case "+":
                    result = firstOperand + secondOperand;
                    break;
                case "-":
                    result = firstOperand - secondOperand;
                    break;
                case "*":
                    result = firstOperand * secondOperand;
                    break;
                case "/":
                    if (secondOperand != 0) {
                        result = firstOperand / secondOperand;
                    } else {
                        Toast.makeText(this, "Cannot divide by zero",
Toast.LENGTH_SHORT).show();
                        return;
                    }
                    break;
            }

            display.setText(String.valueOf(result));
            input = String.valueOf(result);
            operator = "";
            firstOperand = result;
        }
    }
}
```

Slip 3
Q.1) Create an application that allows the user to enter a number in the textbox.Check whether the number in the textbox is Armstrong or not. Print the message accordingly in the label control
----activity_main.xml----

```
<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:padding="16dp"
```

```
    tools:context=".MainActivity">

    <EditText
        android:id="@+id/edtNumber"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter a number"
        android:inputType="number" />

    <Button
        android:id="@+id/btnCheck"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Check Armstrong"
        android:layout_below="@id/edtNumber"
        android:layout_marginTop="16dp" />

    <TextView
        android:id="@+id/tvResult"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text=""
        android:textSize="18sp"
        android:layout_below="@id/btnCheck"
        android:layout_marginTop="16dp" />

</RelativeLayout>
```

----MainActivity.java----

```
package com.example.slip3q1;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.TextView;
import android.widget.Toast;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    EditText edtNumber;
    Button btnCheck;
    TextView tvResult;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        edtNumber = findViewById(R.id.edtNumber);
        btnCheck = findViewById(R.id.btnCheck);
        tvResult = findViewById(R.id.tvResult);

        btnCheck.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                String input = edtNumber.getText().toString();

                if (input.isEmpty()) {
                    Toast.makeText(MainActivity.this, "Please enter a number",
Toast.LENGTH_SHORT).show();
                    return;
                }
```

```
                try {
                    int number = Integer.parseInt(input);
                    if (isArmstrong(number)) {
                        tvResult.setText(number + " is an Armstrong number!");
                    } else {
                        tvResult.setText(number + " is not an Armstrong number.");
                    }
                } catch (NumberFormatException e) {
                    Toast.makeText(MainActivity.this, "Please enter a valid number",
Toast.LENGTH_SHORT).show();
                }
            }
        });
    }

    private boolean isArmstrong(int number) {
        int originalNumber = number;
        int sum = 0;
        int digitsCount = String.valueOf(number).length();
        while (number > 0) {
            int digit = number % 10;
            sum += Math.pow(digit, digitsCount);
            number /= 10;
        }
        return sum == originalNumber;
    }
}
```

Q.2) Create an Android application which examine a phone number entered by a user with the given format.

â€¢ Area code should be one of the following: 040, 041, 050, 0400, 044

â€¢ There should 6 - 8 numbers in telephone number (+ area code).

----activity_main.xml----

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="16dp"
    android:gravity="center">

    <EditText
        android:id="@+id/edtPhoneNumber"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:hint="Enter phone number"
        android:inputType="phone"/>

    <Button
        android:id="@+id/btnCheck"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Check Format"/>

    <TextView
        android:id="@+id/tvResult"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text=""
        android:textSize="18sp"
        android:layout_marginTop="20dp"/>
</LinearLayout>
```

----MainActivity.java----

```
package com.example.slip3q2;

import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.TextView;
import android.widget.Toast;

import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    EditText edtPhoneNumber;
    Button btnCheck;
    TextView tvResult;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        edtPhoneNumber = findViewById(R.id.edtPhoneNumber);
        btnCheck = findViewById(R.id.btnCheck);
        tvResult = findViewById(R.id.tvResult);

        btnCheck.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                String phoneNumber = edtPhoneNumber.getText().toString().trim();

                if (isValidPhoneNumber(phoneNumber)) {
                    tvResult.setText("Valid phone number");
                } else {
                    tvResult.setText("Invalid phone number");
                    Toast.makeText(MainActivity.this, "Please enter a valid phone number.",
Toast.LENGTH_SHORT).show();
                }
            }
        });
    }
    private boolean isValidPhoneNumber(String phoneNumber) {
        return (phoneNumber.matches("^(040|041|050|0400|044)\\d{6,8}$"));
    }
}
```

Slip 5
Q.1) Java Android Program to Demonstrate Alert Dialog Box.

----------- activity_main.xml ---------

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center"
    android:orientation="vertical">

    <Button
        android:id="@+id/showDialogButton"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
```

```
        android:text="Show Alert Dialog" />

</LinearLayout>
```

--------- MainActivity.java ----------

```
package com.example.slip5_q1;

import android.content.DialogInterface;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.Toast;
import androidx.appcompat.app.AlertDialog;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        Button showDialogButton = findViewById(R.id.showDialogButton);

        showDialogButton.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                showAlertDialog();
            }
        });
    }

    private void showAlertDialog() {
        AlertDialog.Builder builder = new AlertDialog.Builder(this);
        builder.setTitle("Alert Dialog Example");
        builder.setMessage("Are you sure you want to proceed?");

        builder.setPositiveButton("OK", new DialogInterface.OnClickListener() {
            @Override
            public void onClick(DialogInterface dialog, int which) {
                Toast.makeText(MainActivity.this, "You clicked OK",
Toast.LENGTH_SHORT).show();
            }
        });

        builder.setNegativeButton("Cancel", new DialogInterface.OnClickListener() {
            @Override
            public void onClick(DialogInterface dialog, int which) {
                Toast.makeText(MainActivity.this, "You clicked Cancel",
Toast.LENGTH_SHORT).show();
                dialog.dismiss();
            }
        });

        builder.setNeutralButton("Retry", new DialogInterface.OnClickListener() {
            @Override
            public void onClick(DialogInterface dialog, int which) {
                Toast.makeText(MainActivity.this, "You clicked Retry",
Toast.LENGTH_SHORT).show();
            }
        });

        AlertDialog alertDialog = builder.create();
```

```
        alertDialog.show();
    }
}
```

Q.2) Create an Android application which will ask the user to input his / her name.A message should display the two items concatenated in a label. Change the formatof the label using radio buttons and check boxes for selection. The user can make the label text bold, underlined or italic as well as change its color. Also include buttons to display the message in the label, clear the text boxes as well as label.Finally exit.

---------------activity_main.xml-----------------

```
<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:padding="16dp">

    <!-- EditText for entering name -->
    <EditText
        android:id="@+id/editTextName"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter your name"
        android:layout_marginTop="50dp"
        android:padding="10dp"
        android:textSize="16sp"/>

    <!-- Button to display message -->
    <Button
        android:id="@+id/buttonDisplay"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Display Message"
        android:layout_below="@id/editTextName"
        android:layout_centerHorizontal="true"
        android:layout_marginTop="20dp"/>

    <!-- Button to clear the text and label -->
    <Button
        android:id="@+id/buttonClear"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Clear"
        android:layout_below="@id/buttonDisplay"
        android:layout_centerHorizontal="true"
        android:layout_marginTop="20dp"/>

    <!-- Button to exit the app -->
    <Button
        android:id="@+id/buttonExit"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Exit"
        android:layout_below="@id/buttonClear"
        android:layout_centerHorizontal="true"
        android:layout_marginTop="20dp"/>

    <!-- TextView for displaying the message -->
    <TextView
        android:id="@+id/textViewMessage"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Hello!"
```

```
        android:textSize="18sp"
        android:layout_below="@id/buttonExit"
        android:layout_centerHorizontal="true"
        android:layout_marginTop="20dp"/>

    <!-- RadioGroup for text formatting -->
    <RadioGroup
        android:id="@+id/radioGroupFormat"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_below="@id/textViewMessage"
        android:layout_marginTop="20dp"
        android:orientation="horizontal">

        <!-- Bold RadioButton -->
        <RadioButton
            android:id="@+id/radioBold"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Bold"/>

        <!-- Italic RadioButton -->
        <RadioButton
            android:id="@+id/radioItalic"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Italic"/>

        <!-- Underline RadioButton -->
        <RadioButton
            android:id="@+id/radioUnderline"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Underline"/>
    </RadioGroup>

    <!-- CheckBox for changing text color -->
    <CheckBox
        android:id="@+id/checkBoxRed"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Red"
        android:layout_below="@id/radioGroupFormat"
        android:layout_marginTop="20dp"
        android:layout_alignParentLeft="true"/>

    <CheckBox
        android:id="@+id/checkBoxBlue"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Blue"
        android:layout_below="@id/radioGroupFormat"
        android:layout_marginTop="20dp"
        android:layout_alignParentRight="true"/>
</RelativeLayout>

-------------------MainActivity.java---------------------

package com.example.slip12_q2;

import android.graphics.Color;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.CheckBox;
```

```
import android.widget.EditText;
import android.widget.RadioButton;
import android.widget.RadioGroup;
import android.widget.TextView;
import android.widget.Toast;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    private EditText editTextName;
    private Button buttonDisplay, buttonClear, buttonExit;
    private TextView textViewMessage;
    private RadioGroup radioGroupFormat;
    private RadioButton radioBold, radioItalic, radioUnderline;
    private CheckBox checkBoxRed, checkBoxBlue;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        // Initialize the views
        editTextName = findViewById(R.id.editTextName);
        buttonDisplay = findViewById(R.id.buttonDisplay);
        buttonClear = findViewById(R.id.buttonClear);
        buttonExit = findViewById(R.id.buttonExit);
        textViewMessage = findViewById(R.id.textViewMessage);
        radioGroupFormat = findViewById(R.id.radioGroupFormat);
        radioBold = findViewById(R.id.radioBold);
        radioItalic = findViewById(R.id.radioItalic);
        radioUnderline = findViewById(R.id.radioUnderline);
        checkBoxRed = findViewById(R.id.checkBoxRed);
        checkBoxBlue = findViewById(R.id.checkBoxBlue);

        // Set the Display button's click listener
        buttonDisplay.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                String name = editTextName.getText().toString().trim();
                if (name.isEmpty()) {
                    Toast.makeText(MainActivity.this, "Please enter your name",
Toast.LENGTH_SHORT).show();
                    return;
                }

                // Create the message
                String message = "Hello, " + name + "!";

                // Apply the selected text formatting
                if (radioBold.isChecked()) {
                    textViewMessage.setTypeface(null, android.graphics.Typeface.BOLD);
                } else if (radioItalic.isChecked()) {
                    textViewMessage.setTypeface(null, android.graphics.Typeface.ITALIC);
                } else if (radioUnderline.isChecked()) {
                    textViewMessage.setPaintFlags(textViewMessage.getPaintFlags() |
android.graphics.Paint.UNDERLINE_TEXT_FLAG);
                } else {
                    textViewMessage.setTypeface(null, android.graphics.Typeface.NORMAL);
                    textViewMessage.setPaintFlags(textViewMessage.getPaintFlags() &
(~android.graphics.Paint.UNDERLINE_TEXT_FLAG));
                }

                // Apply color formatting based on checkbox selection
                if (checkBoxRed.isChecked()) {
                    textViewMessage.setTextColor(Color.RED);
```

```
            } else if (checkBoxBlue.isChecked()) {
                textViewMessage.setTextColor(Color.BLUE);
            } else {
                textViewMessage.setTextColor(Color.BLACK);  // Default color
            }

            // Set the message in the TextView
            textViewMessage.setText(message);
        }
    });

    // Set the Clear button's click listener
    buttonClear.setOnClickListener(new View.OnClickListener() {
        @Override
        public void onClick(View v) {
            // Clear input fields and reset message
            editTextName.setText("");
            textViewMessage.setText("Hello!");
            radioGroupFormat.clearCheck();
            checkBoxRed.setChecked(false);
            checkBoxBlue.setChecked(false);
        }
    });

    // Set the Exit button's click listener
    buttonExit.setOnClickListener(new View.OnClickListener() {
        @Override
        public void onClick(View v) {
            // Exit the application
            finish();
        }
    });
    }
}
```

Slip 7
Q.1] Java Android Program to Demonstrate ProgressBar

--------- activity_main.xml ----------

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="16dp">

    <ProgressBar
        android:id="@+id/progressBar"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:max="100"
        style="?android:attr/progressBarStyleHorizontal" />

    <TextView
        android:id="@+id/progressText"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Progress: 0%"
        android:layout_marginTop="16dp" />

    <Button
        android:id="@+id/startButton"
        android:layout_width="wrap_content"
```

```
        android:layout_height="wrap_content"
        android:text="Start"
        android:layout_marginTop="16dp" />
</LinearLayout>

----MainActivity.java----
package com.example.progressbardemo;

import android.os.Bundle;
import android.os.Handler;
import android.view.View;
import android.widget.Button;
import android.widget.ProgressBar;
import android.widget.TextView;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {
    private ProgressBar progressBar;
    private Button startButton;
    private TextView progressText;
    private Handler handler = new Handler();
    private int progressStatus = 0;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        progressBar = findViewById(R.id.progressBar);
        startButton = findViewById(R.id.startButton);
        progressText = findViewById(R.id.progressText);

        startButton.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                progressStatus = 0;
                progressBar.setProgress(progressStatus);
                progressText.setText("Progress: 0%");
                new Thread(new Runnable() {
                    @Override
                    public void run() {
                        while (progressStatus < 100) {
                            progressStatus += 10;
                            handler.post(new Runnable() {
                                @Override
                                public void run() {
                                    progressBar.setProgress(progressStatus);
                                    progressText.setText("Progress: " + progressStatus +
"%");
                                }
                            });
                            try {
                                Thread.sleep(500);
                            } catch (InterruptedException e) {
                                e.printStackTrace();
                            }
                        }
                    }
                }).start();
            }
        });
    }
}
```

Q.2] Create table Employee (E_id, name, address, ph_no). Create Application for performing the following operation on the table. (Using SQLite database).
i]Insert record of 5 new Employees.
ii]Show all the details of Employee.

```
--------- activity_main.xml ------------
<?xml version="1.0"  encoding="utf-8"?>
<LinearLayout
     xmlns:android="https://schemas.android.com/apk/res/android"
     xmlns:android="https://schemas.android.com/apk/res-auto"
     xmlns:tools="https://schemas.android.com/tools"
     android:layout_width="match_parent"
     android:layout_height="match_parent"
     android:orientation="vertical"
     tools:context=".MainActivity">

<EditText
        android:layout_width="300dp"
        android:layout_height="wrap_content"
        android:id="@+id/et1"
        android:hint="Enter ID"/>

<EditText
        android:layout_width="300dp"
        android:layout_height="wrap_content"
        android:id="@+id/et2"
        android:hint="Enter Name"/>

<EditText
        android:layout_width="300dp"
        android:layout_height="wrap_content"
        android:id="@+id/et3"
        android:hint="Enter Address"/>

<EditText
        android:layout_width="300dp"
        android:layout_height="wrap_content"
        android:id="@+id/et4"
        android:hint="Enter Phone No"/>

<Button
        android:layout_width="300dp"
        android:layout_height="300dp"
        android:id="@+id/bt1"
        android:hint="Save"/>

<Button
        android:layout_width="300dp"
        android:layout_height="300dp"
        android:id="@+id/bt"
        android:hint="Show"/>

<ListView
        android:layout_width="match_parent"
        android:layout_height="match_parent"
        android:id="@+id/lv"/>

</LinearLayout>


--------- MainActivity.java ------------
package com.example.database;

import androidx.appcompat.app.AppCompatActivity;

import android.os.Bundle;
```

```
import android.view.View;
import android.widget.ArrayAdapter;
import android.widget.Button;
import android.widget.EditText;
import android.widget.ListView;
import android.widget.Toast;

import java.util.ArrayList;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        mydb db = new mydb(MainActivity.this);
        ListView l = (ListView) findViewById(R.id.lv);
        EditText eeid =(EditText) findViewById(R.id.et1);
        EditText eename =(EditText) findViewById(R.id.et2);
        EditText eeadd =(EditText) findViewById(R.id.et3);
        EditText eecno =(EditText) findViewById(R.id.et4);
        Button b = (Button) findViewById(R.id.bt1);
        Button b2 = (Button) findViewById(R.id.bt2);
        b.setOnClickListener(new View.OnClickListener()
        {
            @Override
            public void onClick(View v)
            {
                final String id = eeid.getText().toString();
                final String name = eename.getText().toString();
                final String add = eeadd.getText().toString();
                final String pno = eecno.getText().toString();
                long retValue = db.addDetails(id,name,add,pno);
                if(retValue >= 0)
                    Toast.makeText(getApplicationContext(),"Data Saved
Successfully",Toast.LENGTH_LONG).show();
                else
                    Toast.makeText(getApplicationContext(),"Unable to save
data",Toast.LENGTH_LONG).show();
            }
        });
        b2.setOnClickListener(new View.OnClickListener()
        {
            @Override
            public void onClick(View v)
            {
                String s="";
                ArrayList<String> c=db.getDetails(getApplicationContext());
                /*Iterator itr = c.iterator();
                while(itr.hasNext())
                {

Toast.makeText(getApplicationContext(),itr.next().toString(),Toast.LENGTH_LONG).show();
                 */
                ArrayAdapter<String> aa = new ArrayAdapter<String>
(getApplicationContext(),android.R.layout.simple_list_item_1,c);
                l.setAdapter(aa);
            }
        });
    }
}



------------ mydb.java -------------
```

```
package com.example.database;

import android.content.ContentValues;
import android.content.Context;
import android.database.Cursor;
import android.database.sqlite.SQLiteDatabase;
import android.database.sqlite.SQLiteOpenHelper;

import java.util.ArrayList;
import java.util.Collections;

public class mydb<customer> extends SQLiteOpenHelper
{
    static String dbname="student.db";
    public static final int dbversion=1;
    public mydb(Context c)
    {
        super(c,dbname,null,dbversion);
    }
    public mydb(Context c, String name, SQLiteDatabase.CursorFactory factory,int version)
    {
        super(c,name,factory,version);
    }
    @Override
    public void onCreate(SQLiteDatabase dba)
    {
        dba.execSQL("create table employee(eid integer PRIMARY KEY,ename text,eadd text,epno
text)");
    }
    @Override
    public void onUpgrade(SQLiteDatabase db,int oidVersion,int newVersion)
    {

    }
    public ArrayList<String> getDetails(Context c11)
    {
        ArrayList<String> s1 = new ArrayList<String>(Collections.singleton(""));
        String sql = "select * from employee";
        SQLiteDatabase db = this.getReadableDatabase();
        Cursor c = db.rawQuery(sql,null);
        if(c.getCount() > 0)
        {
            c.moveToFirst(); do
            {
                Integer id = c.getString(c.getColumnIndexOrThrow("eid"));
                String name = c.getString(c.getColumnIndexOrThrow("ename"));
                String addr = c.getString(c.getColumnIndexOrThrow("eadd "));
                String phno = c.getString(c.getColumnIndexOrThrow("epno"));
                String c1 ="Emp Id : "+ id  + "\n" "Name : "+ name + "\n "+"Address : " +
addr+ " \n"+ "Phone No. : "+ phno+"\n";
                s1.add(c1);
            }
            while(c.moveToNext());
        }
        return s1;
    }
    public long addDetails(Integer id,String name,String addr,String no)
    {
        ContentValues cv = new ContentValues();
        cv.put("eid",id);
        cv.put("ename",name);
        cv.put("eadd",addr);
        cv.put("epno",no);
        SQLiteDatabase db = this.getWritableDatabase();
        long retValue = db.insert("employee",null,cv);
```

```
        return retValue;
    }
}
```

```
----------- AndroidManifest.xml --------------
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools">

    <application
        android:allowBackup="true"
        android:dataExtractionRules="@xml/data_extraction_rules"
        android:fullBackupContent="@xml/backup_rules"
        android:icon="@mipmap/ic_launcher"
        android:label="@string/app_name"
        android:roundIcon="@mipmap/ic_launcher_round"
        android:supportsRtl="true"
        android:theme="@style/Theme.Database"
        tools:targetApi="31">
        <activity
            android:name=".MainActivity"
            android:exported="true">
            <intent-filter>
                <action android:name="android.intent.action.MAIN" />

                <category android:name="android.intent.category.LAUNCHER" />
            </intent-filter>
        </activity>
    </application>

</manifest>
```

Slip 8
Q.1] Create a Application which shows Life Cycle of Activity.

acivity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity"/>
```

-------------MainActivity.java------

```
package com.example.myapplication12;

import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.widget.Toast;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
    }
    protected void onStart()
    {
```

```
        Toast.makeText(this, "welcome", Toast.LENGTH_SHORT).show();
        super.onStart();
    }

    @Override
    protected void onResume()
    {
        Toast.makeText(this, "onResume() called", Toast.LENGTH_SHORT).show();
        super.onResume();
    }

    protected void onPause()
    {
        Toast.makeText(this, "onPause() called",Toast.LENGTH_SHORT).show();
        super.onPause();
    }
    protected void onStop()
    {
        Toast.makeText(this, "onStop() called",Toast.LENGTH_SHORT).show();
        super.onStop();
    }
    protected void onDestroy()
    {
        Toast.makeText(this, "onDestroy() called",Toast.LENGTH_SHORT).show();
        super.onDestroy();
    }
}
```

Q.2] Create table Customer (id, name, address, ph_no). Create Application for performing the following operation on the table. (Using SQLite database).
i]Insert new customer details (At least 5 records).
ii]Show all the customer details.

```
--------- activity_main.xml ------------
<?xml version="1.0"  encoding="utf-8"?>
<LinearLayout
     xmlns:android="https://schemas.android.com/apk/res/android"
     xmlns:android="https://schemas.android.com/apk/res-auto"
     xmlns:tools="https://schemas.android.com/tools"
     android:layout_width="match_parent"
     android:layout_height="match_parent"
     android:orientation="vertical"
     tools:context=".MainActivity">

<EditText
        android:layout_width="300dp"
        android:layout_height="wrap_content"
        android:id="@+id/et1"
        android:hint="Enter Id"/>

<EditText
        android:layout_width="300dp"
        android:layout_height="wrap_content"
        android:id="@+id/et2"
        android:hint="Enter Name"/>

<EditText
        android:layout_width="300dp"
        android:layout_height="wrap_content"
        android:id="@+id/et3"
        android:hint="Enter Address"/>

<EditText
        android:layout_width="300dp"
        android:layout_height="wrap_content"
```

```
        android:id="@+id/et4"
        android:hint="Enter Contact Number"/>

<Button
        android:layout_width="300dp"
        android:layout_height="300dp"
        android:id="@+id/bt1"
        android:hint="Save"/>

<Button
        android:layout_width="300dp"
        android:layout_height="300dp"
        android:id="@+id/bt"
        android:hint="Show"/>

<ListView
        android:layout_width="match_parent"
        android:layout_height="match_parent"
        android:id="@+id/lv"/>

</LinearLayout>


--------- MainActivity.java ------------
package com.example.database;

import androidx.appcompat.app.AppCompatActivity;

import android.os.Bundle;
import android.view.View;
import android.widget.ArrayAdapter;
import android.widget.Button;
import android.widget.EditText;
import android.widget.ListView;
import android.widget.Toast;

import java.util.ArrayList;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        mydb db = new mydb(MainActivity.this);
        ListView l = (ListView) findViewById(R.id.lv);
        EditText ecid =(EditText) findViewById(R.id.et1);
        EditText ecname =(EditText) findViewById(R.id.et2);
        EditText ecno =(EditText) findViewById(R.id.et3);
        EditText ecadd =(EditText) findViewById(R.id.et4);
        Button b = (Button) findViewById(R.id.bt1);
        Button b2 = (Button) findViewById(R.id.bt2);
        b.setOnClickListener(new View.OnClickListener()
        {
            @Override
            public void onClick(View v)
            {
                final String id = ecid.getText().toString();
                final String name = ecname.getText().toString();
                final String no = ecno.getText().toString();
                final String add = ecadd.getText().toString();
                long retValue = db.addDetails(id,name,no,add);
                if(retValue >= 0)
                    Toast.makeText(getApplicationContext(),"Data Saved
Successfully",Toast.LENGTH_LONG).show();
```

```
                else
                    Toast.makeText(getApplicationContext(),"Unable to save
data",Toast.LENGTH_LONG).show();
            }
        });
        b2.setOnClickListener(new View.OnClickListener()
        {
            @Override
            public void onClick(View v)
            {
                String s="";
                ArrayList<String> c=db.getDetails(getApplicationContext());
                /*Iterator itr = c.iterator();
                while(itr.hasNext())
                {

Toast.makeText(getApplicationContext(),itr.next().toString(),Toast.LENGTH_LONG).show();
                 */
                ArrayAdapter<String> aa = new ArrayAdapter<String>
(getApplicationContext(),android.R.layout.simple_list_item_1,c);
                l.setAdapter(aa);
            }
        });
    }
}
```

```
------------ mydb.java -------------
package com.example.database;

import android.content.ContentValues;
import android.content.Context;
import android.database.Cursor;
import android.database.sqlite.SQLiteDatabase;
import android.database.sqlite.SQLiteOpenHelper;

import java.util.ArrayList;
import java.util.Collections;

public class mydb<customer> extends SQLiteOpenHelper
{
    static String dbname="customer.db";
    public static final int dbversion=1;
    public mydb(Context c)
    {
        super(c,dbname,null,dbversion);
    }
    public mydb(Context c, String name, SQLiteDatabase.CursorFactory factory,int version)
    {
        super(c,name,factory,version);
    }
    @Override
    public void onCreate(SQLiteDatabase dba)
    {
        dba.execSQL("create table customer(cid integer PRIMARY KEY,cname text,cphone
text,cadd text)");
    }
    @Override
    public void onUpgrade(SQLiteDatabase db,int oidVersion,int newVersion)
    {

    }
    public ArrayList<String> getDetails(Context c11)
    {
```

```
        ArrayList<String> s1 = new ArrayList<String>(Collections.singleton(""));
        String sql = "select * from customer";
        SQLiteDatabase db = this.getReadableDatabase();
        Cursor c = db.rawQuery(sql,null);
        if(c.getCount() > 0)
        {
            c.moveToFirst(); do
            {
                Integer id = c.getString(c.getColumnIndexOrThrow("cid"));
                String name = c.getString(c.getColumnIndexOrThrow("cname"));
                String mob = c.getString(c.getColumnIndexOrThrow("cphone"));
                String addr = c.getString(c.getColumnIndexOrThrow("cadd"));
                String c1 ="ID : "+ id + "\n" + "Name : "+ name + "\n "+"Address : " + addr+
" \n"+ "Contact No. : "+ mob+"\n";
                s1.add(c1);
            }
            while(c.moveToNext());
        }
        return s1;
    }
    public long addDetails(Integer id,String name,String no,String addr)
    {
        ContentValues cv = new ContentValues();
        cv.put("cid",id);
        cv.put("cname",name);
        cv.put("cphone",no);
        cv.put("cadd",addr);
        SQLiteDatabase db = this.getWritableDatabase();
        long retValue = db.insert("customer",null,cv);
        return retValue;
    }
}


----------- AndroidManifest.xml --------------
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools">

    <application
        android:allowBackup="true"
        android:dataExtractionRules="@xml/data_extraction_rules"
        android:fullBackupContent="@xml/backup_rules"
        android:icon="@mipmap/ic_launcher"
        android:label="@string/app_name"
        android:roundIcon="@mipmap/ic_launcher_round"
        android:supportsRtl="true"
        android:theme="@style/Theme.Database"
        tools:targetApi="31">
        <activity
            android:name=".MainActivity"
            android:exported="true">
            <intent-filter>
                <action android:name="android.intent.action.MAIN" />

                <category android:name="android.intent.category.LAUNCHER" />
            </intent-filter>
        </activity>
    </application>

</manifest>
```

Slip 9

Q.1] Create an application that allows the user to enter a number in the textbox named â€žgetnumâ€Ÿ. Check whether the number in the textbox â€žgetnumâ€Ÿ is Palindrome or not. Print the message accordingly in the label when the user clicks on the button â€žCheckâ€Ÿ.

---------------activity_main.xml-----------------

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <EditText
        android:id="@+id/getnum"
        android:layout_width="0dp"
        android:layout_height="wrap_content"
        android:hint="Enter a number"
        android:inputType="number"
        android:layout_marginTop="100dp"
        android:layout_marginStart="32dp"
        android:layout_marginEnd="32dp"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintEnd_toEndOf="parent"/>

    <Button
        android:id="@+id/checkButton"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Check"
        android:onClick="checkPalindrome"
        app:layout_constraintTop_toBottomOf="@id/getnum"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        android:layout_marginTop="30dp"/>

    <TextView
        android:id="@+id/resultLabel"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text=""
        android:textSize="18sp"
        app:layout_constraintTop_toBottomOf="@id/checkButton"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        android:layout_marginTop="40dp"/>

</androidx.constraintlayout.widget.ConstraintLayout>
```

---------------------ActivityMain.java-------------------------

```
package com.example.palindrome;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.TextView;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    private EditText getnum;
```

```
    private Button checkButton;
    private TextView resultLabel;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        // Initialize the views
        getnum = findViewById(R.id.getnum);
        checkButton = findViewById(R.id.checkButton);
        resultLabel = findViewById(R.id.resultLabel);
    }
    public void checkPalindrome(View view) {
        String input = getnum.getText().toString().trim();

        if (input.isEmpty()) {
            resultLabel.setText("Please enter a valid number.");

resultLabel.setTextColor(getResources().getColor(android.R.color.holo_red_dark));
            return;
        }
        String reversedInput = new StringBuilder(input).reverse().toString();
        if (input.equals(reversedInput)) {
            resultLabel.setText("The number is a palindrome!");

resultLabel.setTextColor(getResources().getColor(android.R.color.holo_green_dark));
        } else {
            resultLabel.setText("The number is not a palindrome.");

resultLabel.setTextColor(getResources().getColor(android.R.color.holo_red_dark));
        }
    }
}
```

Q.2] Java android program to create simple calculator.

-------------------------activity_main.xml-----------------------------

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <!-- Display TextView for the Calculator -->
    <TextView
        android:id="@+id/display"
        android:layout_width="0dp"
        android:layout_height="wrap_content"
        android:text=""
        android:textSize="36sp"
        android:gravity="end"
        android:padding="20dp"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintBottom_toTopOf="@+id/gridLayout"/>
```

```xml
    <!-- Grid Layout for calculator buttons -->
    <GridLayout
        android:id="@+id/gridLayout"
        android:layout_width="0dp"
        android:layout_height="0dp"
        android:columnCount="4"
        android:rowCount="5"
        app:layout_constraintTop_toBottomOf="@id/display"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintBottom_toBottomOf="parent">

        <!-- First Row -->
        <Button android:id="@+id/button1" android:text="1"/>
        <Button android:id="@+id/button2" android:text="2"/>
        <Button android:id="@+id/button3" android:text="3"/>
        <Button android:id="@+id/buttonDivide" android:text="/"/>

        <!-- Second Row -->
        <Button android:id="@+id/button4" android:text="4"/>
        <Button android:id="@+id/button5" android:text="5"/>
        <Button android:id="@+id/button6" android:text="6"/>
        <Button android:id="@+id/buttonMultiply" android:text="*"/>

        <!-- Third Row -->
        <Button android:id="@+id/button7" android:text="7"/>
        <Button android:id="@+id/button8" android:text="8"/>
        <Button android:id="@+id/button9" android:text="9"/>
        <Button android:id="@+id/buttonSubtract" android:text="-"/>

        <!-- Fourth Row -->
        <Button android:id="@+id/button0" android:text="0"/>
        <Button android:id="@+id/buttonClear" android:text="C"/>
        <Button android:id="@+id/buttonEqual" android:text="="/>
        <Button android:id="@+id/buttonAdd" android:text="+"/>

    </GridLayout>

</androidx.constraintlayout.widget.ConstraintLayout>
```

-------------------MainActivity.java---------------------

```java
package com.example.slip9q2;

import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.TextView;
import android.widget.Toast;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    private TextView display;
    private String input = "";
    private String operator = "";
    private double firstOperand = 0;
    private double secondOperand = 0;
    private boolean operatorClicked = false;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
```

```
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        display = findViewById(R.id.display);


        Button button0 = findViewById(R.id.button0);
        Button button1 = findViewById(R.id.button1);
        Button button2 = findViewById(R.id.button2);
        Button button3 = findViewById(R.id.button3);
        Button button4 = findViewById(R.id.button4);
        Button button5 = findViewById(R.id.button5);
        Button button6 = findViewById(R.id.button6);
        Button button7 = findViewById(R.id.button7);
        Button button8 = findViewById(R.id.button8);
        Button button9 = findViewById(R.id.button9);

        Button buttonAdd = findViewById(R.id.buttonAdd);
        Button buttonSubtract = findViewById(R.id.buttonSubtract);
        Button buttonMultiply = findViewById(R.id.buttonMultiply);
        Button buttonDivide = findViewById(R.id.buttonDivide);
        Button buttonClear = findViewById(R.id.buttonClear);
        Button buttonEqual = findViewById(R.id.buttonEqual);


        button0.setOnClickListener(v -> appendToInput("0"));
        button1.setOnClickListener(v -> appendToInput("1"));
        button2.setOnClickListener(v -> appendToInput("2"));
        button3.setOnClickListener(v -> appendToInput("3"));
        button4.setOnClickListener(v -> appendToInput("4"));
        button5.setOnClickListener(v -> appendToInput("5"));
        button6.setOnClickListener(v -> appendToInput("6"));
        button7.setOnClickListener(v -> appendToInput("7"));
        button8.setOnClickListener(v -> appendToInput("8"));
        button9.setOnClickListener(v -> appendToInput("9"));
        buttonAdd.setOnClickListener(v -> setOperator("+"));
        buttonSubtract.setOnClickListener(v -> setOperator("-"));
        buttonMultiply.setOnClickListener(v -> setOperator("*"));
        buttonDivide.setOnClickListener(v -> setOperator("/"));
        buttonClear.setOnClickListener(v -> clearInput());

        buttonEqual.setOnClickListener(v -> calculateResult());
    }

    private void appendToInput(String value) {
        if (operatorClicked) {
            input = "";
            operatorClicked = false;
        }
        input += value;
        display.setText(input);
    }

    private void setOperator(String operator) {
        if (!input.isEmpty()) {
            firstOperand = Double.parseDouble(input);
            this.operator = operator;
            input = "";
            operatorClicked = true;
        }
    }

    private void clearInput() {
        input = "";
        display.setText(input);
```

```
        firstOperand = 0;
        secondOperand = 0;
        operator = "";
    }

    private void calculateResult() {
        if (!input.isEmpty()) {
            secondOperand = Double.parseDouble(input);
            double result = 0;

            switch (operator) {
                case "+":
                    result = firstOperand + secondOperand;
                    break;
                case "-":
                    result = firstOperand - secondOperand;
                    break;
                case "*":
                    result = firstOperand * secondOperand;
                    break;
                case "/":
                    if (secondOperand != 0) {
                        result = firstOperand / secondOperand;
                    } else {
                        Toast.makeText(this, "Cannot divide by zero",
Toast.LENGTH_SHORT).show();
                        return;
                    }
                    break;
            }

            display.setText(String.valueOf(result));
            input = String.valueOf(result);
            operator = "";
            firstOperand = result;
        }
    }
}
```

Slip 11
Q.1] Create an Android Application to accept two numbers to calculate its Power and Average. Create two buttons: Power and Average. Display the appropriate result on the next activity on Button click.

```
---------------activity_main.xml-----------------
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">


        <!-- EditText for First Number -->
        <EditText
            android:id="@+id/editTextNumber1"
            android:layout_width="0dp"
            android:layout_height="wrap_content"
            android:hint="Enter first number"
            android:inputType="numberDecimal"
            app:layout_constraintEnd_toEndOf="parent"
            app:layout_constraintStart_toStartOf="parent"
```

```
            app:layout_constraintTop_toTopOf="parent"
            android:padding="16dp" />

        <!-- EditText for Second Number -->
        <EditText
            android:id="@+id/editTextNumber2"
            android:layout_width="0dp"
            android:layout_height="wrap_content"
            android:hint="Enter second number"
            android:inputType="numberDecimal"
            app:layout_constraintEnd_toEndOf="parent"
            app:layout_constraintStart_toStartOf="parent"
            app:layout_constraintTop_toBottomOf="@id/editTextNumber1"
            android:padding="16dp" />

        <!-- Button for Calculating Power -->
        <Button
            android:id="@+id/buttonPower"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Power"
            app:layout_constraintEnd_toEndOf="parent"
            app:layout_constraintStart_toStartOf="parent"
            app:layout_constraintTop_toBottomOf="@id/editTextNumber2"
            android:layout_marginTop="16dp"/>

        <!-- Button for Calculating Average -->
        <Button
            android:id="@+id/buttonAverage"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Average"
            app:layout_constraintEnd_toEndOf="parent"
            app:layout_constraintStart_toStartOf="parent"
            app:layout_constraintTop_toBottomOf="@id/buttonPower"
            android:layout_marginTop="16dp"/>

        <!-- TextView for displaying the result -->
        <TextView
            android:id="@+id/resultTextView"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Result will appear here"
            android:textSize="18sp"
            app:layout_constraintEnd_toEndOf="parent"
            app:layout_constraintStart_toStartOf="parent"
            app:layout_constraintTop_toBottomOf="@id/buttonAverage"
            android:padding="16dp" />

</androidx.constraintlayout.widget.ConstraintLayout>

-------------------MainActivity.java---------------------

package com.example.slip11q1;

        import android.content.Intent;
        import android.os.Bundle;
        import android.view.View;
        import android.widget.Button;
        import android.widget.EditText;
        import android.widget.TextView;
        import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {
```

```
    EditText editTextNumber1, editTextNumber2;
    Button buttonPower, buttonAverage;
    TextView resultTextView;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        // Initialize the views
        editTextNumber1 = findViewById(R.id.editTextNumber1);
        editTextNumber2 = findViewById(R.id.editTextNumber2);
        buttonPower = findViewById(R.id.buttonPower);
        buttonAverage = findViewById(R.id.buttonAverage);
        resultTextView = findViewById(R.id.resultTextView);

        // Power button click listener
        buttonPower.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                // Get input values
                double num1 = Double.parseDouble(editTextNumber1.getText().toString());
                double num2 = Double.parseDouble(editTextNumber2.getText().toString());

                // Calculate power
                double result = Math.pow(num1, num2);

                // Display result
                resultTextView.setText("Power Result: " + result);
            }
        });

        // Average button click listener
        buttonAverage.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                // Get input values
                double num1 = Double.parseDouble(editTextNumber1.getText().toString());
                double num2 = Double.parseDouble(editTextNumber2.getText().toString());

                // Calculate average
                double result = (num1 + num2) / 2;

                // Display result
                resultTextView.setText("Average Result: " + result);
            }
        });
    }
}
```

Q.2] Create an Android Application to perform following string operation according to user selection of radio button.

```
---------------activity_main.xml-----------------
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

        <!-- Instruction Text -->
        <TextView
            android:id="@+id/instructionText"
```

```xml
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Enter String:"
    android:textSize="18sp"
    app:layout_constraintBottom_toTopOf="@id/editText"
    app:layout_constraintLeft_toLeftOf="parent"
    app:layout_constraintTop_toTopOf="parent"
    app:layout_constraintRight_toRightOf="parent"/>

<!-- EditText for user input -->
<EditText
    android:id="@+id/editText"
    android:layout_width="0dp"
    android:layout_height="wrap_content"
    android:hint="Enter string here"
    android:textSize="18sp"
    app:layout_constraintBottom_toTopOf="@id/radioGroup"
    app:layout_constraintLeft_toLeftOf="parent"
    app:layout_constraintRight_toRightOf="parent"
    app:layout_constraintTop_toBottomOf="@id/instructionText"/>

<!-- RadioButtons for operations -->
<RadioGroup
    android:id="@+id/radioGroup"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:orientation="vertical"
    app:layout_constraintBottom_toTopOf="@id/performButton"
    app:layout_constraintLeft_toLeftOf="parent"
    app:layout_constraintTop_toBottomOf="@id/editText"
    app:layout_constraintRight_toRightOf="parent">

    <RadioButton
        android:id="@+id/radioReverse"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Reverse String" />

    <RadioButton
        android:id="@+id/radioUppercase"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Uppercase" />

    <RadioButton
        android:id="@+id/radioLowercase"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Lowercase" />

    <RadioButton
        android:id="@+id/radioRightFive"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Right 5 Characters" />

    <RadioButton
        android:id="@+id/radioLeftFive"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Left 5 Characters" />

</RadioGroup>

<!-- Button to perform operation -->
```

```
        <Button
            android:id="@+id/performButton"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Check"
            app:layout_constraintBottom_toTopOf="@id/resultText"
            app:layout_constraintLeft_toLeftOf="parent"
            app:layout_constraintRight_toRightOf="parent"
            app:layout_constraintTop_toBottomOf="@id/radioGroup"/>

        <!-- TextView to display result -->
        <TextView
            android:id="@+id/resultText"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Output:"
            android:textSize="18sp"
            app:layout_constraintBottom_toBottomOf="parent"
            app:layout_constraintLeft_toLeftOf="parent"
            app:layout_constraintTop_toBottomOf="@id/performButton"
            app:layout_constraintRight_toRightOf="parent"/>


</androidx.constraintlayout.widget.ConstraintLayout>
```

-------------------MainActivity.java---------------------

```
package com.example.slip11q2;

import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.RadioButton;
import android.widget.RadioGroup;
import android.widget.TextView;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    private EditText editTextInput;
    private RadioGroup radioGroupOperations;
    private Button buttonPerformOperation;
    private TextView textViewResult;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        // Initialize UI components
        editTextInput = findViewById(R.id.editText);
        radioGroupOperations = findViewById(R.id.radioGroup);
        buttonPerformOperation = findViewById(R.id.performButton);
        textViewResult = findViewById(R.id.resultText);

        // Set a click listener on the button to perform the selected operation
        buttonPerformOperation.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                // Get the input string from EditText
                String inputString = editTextInput.getText().toString().trim();

                if (inputString.isEmpty()) {
```

```
                    textViewResult.setText("Please enter a string.");
                    return;
                }

                // Get selected radio button ID
                int selectedRadioId = radioGroupOperations.getCheckedRadioButtonId();
                String result = "";

                if (selectedRadioId == -1) {
                    // No radio button selected
                    result = "Please select an operation.";
                } else {
                    // Use if-else instead of switch to handle the selected radio button
IDpoonam
                    if (selectedRadioId == R.id.radioReverse) {
                        result = reverseString(inputString);
                    } else if (selectedRadioId == R.id.radioUppercase) {
                        result = inputString.toUpperCase();
                    } else if (selectedRadioId == R.id.radioLowercase) {
                        result = inputString.toLowerCase();
                    }
                    else if (selectedRadioId == R.id.radioRightFive) {
                        // Check if the string has at least 5 characters
                        if (inputString.length() >= 5) {
                            String rightFive = inputString.substring(inputString.length() -
5);

                            result = "Right 5 Characters: " + rightFive;
                        } else {
                            result = "String is too short.";
                        }
                    }
                    else if (selectedRadioId == R.id.radioLeftFive) {
                            // Check if the string has at least 5 characters
                            if (inputString.length() >= 5) {
                                String leftFive = inputString.substring(0, 5);
                                result = "Left 5 Characters: " + leftFive;
                            } else {
                                result = "String is too short.";
                            }
                    }else {
                        result = "Invalid operation.";
                    }
                }

                // Display the result
                textViewResult.setText("Result: " + result);
            }
        });
    }

    // Function to reverse the string
    private String reverseString(String input) {
        StringBuilder reversed = new StringBuilder(input);
        return reversed.reverse().toString();
    }
}
```

Slip 12
Q.1] Construct an Android app that toggles a light bulb ON and OFF when the user clicks on toggle button
---------------activity_main.xml-----------------

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
```

```
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    android:padding="16dp">

    <!-- ImageView for displaying the light bulb -->
    <ImageView
        android:id="@+id/imageViewLightBulb"
        android:layout_width="200dp"
        android:layout_height="200dp"
        android:src="@drawable/bulb_off" />  <!-- Initial image set to bulb_off -->

    <!-- Toggle Button to turn the light on/off -->
    <Button
        android:id="@+id/buttonToggle"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Turn On Light"
        android:layout_marginTop="20dp"/>

</LinearLayout>

-------------------MainActivity.java---------------------
package com.example.slip12_q1;

import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.ImageView;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    private ImageView imageViewLightBulb;
    private Button buttonToggle;
    private boolean isLightOn = false;  // Variable to track the state of the light

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        // Initialize the views
        imageViewLightBulb = findViewById(R.id.imageViewLightBulb);
        buttonToggle = findViewById(R.id.buttonToggle);

        // Set the button click listener to toggle the light bulb state
        buttonToggle.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                // Toggle the state
                isLightOn = !isLightOn;

                // Change the light bulb image based on the state
                if (isLightOn) {
                    imageViewLightBulb.setImageResource(R.drawable.bulp_on);  // Set to ON
image
                    buttonToggle.setText("Turn Off Light");  // Update button text
                } else {
                    imageViewLightBulb.setImageResource(R.drawable.bulb_off);  // Set to OFF
image
                    buttonToggle.setText("Turn On Light");  // Update button text
                }
            }
        });
```

```
    }
}
```

Q.2] Create an Android application which will ask the user to input his / her name. A message should display the two items concatenated in a label. Change the format of the label using radio buttons and check boxes for selection. The user can make the label text bold, underlined or italic as well as change its color. Also include buttons to display the message in the label, clear the text boxes as well as label. Finally exit.

---------------activity_main.xml-----------------

```
<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:padding="16dp">

    <!-- EditText for entering name -->
    <EditText
        android:id="@+id/editTextName"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter your name"
        android:layout_marginTop="50dp"
        android:padding="10dp"
        android:textSize="16sp"/>

    <!-- Button to display message -->
    <Button
        android:id="@+id/buttonDisplay"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Display Message"
        android:layout_below="@id/editTextName"
        android:layout_centerHorizontal="true"
        android:layout_marginTop="20dp"/>

    <!-- Button to clear the text and label -->
    <Button
        android:id="@+id/buttonClear"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Clear"
        android:layout_below="@id/buttonDisplay"
        android:layout_centerHorizontal="true"
        android:layout_marginTop="20dp"/>

    <!-- Button to exit the app -->
    <Button
        android:id="@+id/buttonExit"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Exit"
        android:layout_below="@id/buttonClear"
        android:layout_centerHorizontal="true"
        android:layout_marginTop="20dp"/>

    <!-- TextView for displaying the message -->
    <TextView
        android:id="@+id/textViewMessage"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Hello!"
        android:textSize="18sp"
        android:layout_below="@id/buttonExit"
        android:layout_centerHorizontal="true"
```

```xml
        android:layout_marginTop="20dp"/>

    <!-- RadioGroup for text formatting -->
    <RadioGroup
        android:id="@+id/radioGroupFormat"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_below="@id/textViewMessage"
        android:layout_marginTop="20dp"
        android:orientation="horizontal">

        <!-- Bold RadioButton -->
        <RadioButton
            android:id="@+id/radioBold"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Bold"/>

        <!-- Italic RadioButton -->
        <RadioButton
            android:id="@+id/radioItalic"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Italic"/>

        <!-- Underline RadioButton -->
        <RadioButton
            android:id="@+id/radioUnderline"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Underline"/>
    </RadioGroup>

    <!-- CheckBox for changing text color -->
    <CheckBox
        android:id="@+id/checkBoxRed"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Red"
        android:layout_below="@id/radioGroupFormat"
        android:layout_marginTop="20dp"
        android:layout_alignParentLeft="true"/>

    <CheckBox
        android:id="@+id/checkBoxBlue"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Blue"
        android:layout_below="@id/radioGroupFormat"
        android:layout_marginTop="20dp"
        android:layout_alignParentRight="true"/>
</RelativeLayout>
```

-------------------MainActivity.java---------------------

```java
package com.example.slip12_q2;

import android.graphics.Color;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.CheckBox;
import android.widget.EditText;
import android.widget.RadioButton;
import android.widget.RadioGroup;
```

```
import android.widget.TextView;
import android.widget.Toast;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    private EditText editTextName;
    private Button buttonDisplay, buttonClear, buttonExit;
    private TextView textViewMessage;
    private RadioGroup radioGroupFormat;
    private RadioButton radioBold, radioItalic, radioUnderline;
    private CheckBox checkBoxRed, checkBoxBlue;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        // Initialize the views
        editTextName = findViewById(R.id.editTextName);
        buttonDisplay = findViewById(R.id.buttonDisplay);
        buttonClear = findViewById(R.id.buttonClear);
        buttonExit = findViewById(R.id.buttonExit);
        textViewMessage = findViewById(R.id.textViewMessage);
        radioGroupFormat = findViewById(R.id.radioGroupFormat);
        radioBold = findViewById(R.id.radioBold);
        radioItalic = findViewById(R.id.radioItalic);
        radioUnderline = findViewById(R.id.radioUnderline);
        checkBoxRed = findViewById(R.id.checkBoxRed);
        checkBoxBlue = findViewById(R.id.checkBoxBlue);

        // Set the Display button's click listener
        buttonDisplay.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                String name = editTextName.getText().toString().trim();
                if (name.isEmpty()) {
                    Toast.makeText(MainActivity.this, "Please enter your name",
Toast.LENGTH_SHORT).show();
                    return;
                }

                // Create the message
                String message = "Hello, " + name + "!";

                // Apply the selected text formatting
                if (radioBold.isChecked()) {
                    textViewMessage.setTypeface(null, android.graphics.Typeface.BOLD);
                } else if (radioItalic.isChecked()) {
                    textViewMessage.setTypeface(null, android.graphics.Typeface.ITALIC);
                } else if (radioUnderline.isChecked()) {
                    textViewMessage.setPaintFlags(textViewMessage.getPaintFlags() |
android.graphics.Paint.UNDERLINE_TEXT_FLAG);
                } else {
                    textViewMessage.setTypeface(null, android.graphics.Typeface.NORMAL);
                    textViewMessage.setPaintFlags(textViewMessage.getPaintFlags() &
(~android.graphics.Paint.UNDERLINE_TEXT_FLAG));
                }

                // Apply color formatting based on checkbox selection
                if (checkBoxRed.isChecked()) {
                    textViewMessage.setTextColor(Color.RED);
                } else if (checkBoxBlue.isChecked()) {
                    textViewMessage.setTextColor(Color.BLUE);
                } else {
```

```
                textViewMessage.setTextColor(Color.BLACK);  // Default color
            }

            // Set the message in the TextView
            textViewMessage.setText(message);
        }
    });

    // Set the Clear button's click listener
    buttonClear.setOnClickListener(new View.OnClickListener() {
        @Override
        public void onClick(View v) {
            // Clear input fields and reset message
            editTextName.setText("");
            textViewMessage.setText("Hello!");
            radioGroupFormat.clearCheck();
            checkBoxRed.setChecked(false);
            checkBoxBlue.setChecked(false);
        }
    });

    // Set the Exit button's click listener
    buttonExit.setOnClickListener(new View.OnClickListener() {
        @Override
        public void onClick(View v) {
            // Exit the application
            finish();
        }
    });
  }
}
```

Slip 13
Q.1] Java android program to demonstrate Registration form with validation
---------------activity_main.xml-----------------

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="16dp">

    <EditText
        android:id="@+id/nameInput"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter Name"
        android:minHeight="48dp" />

    <EditText
        android:id="@+id/emailInput"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter Email"
        android:inputType="textEmailAddress"
        android:minHeight="48dp" />

    <EditText
        android:id="@+id/passwordInput"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter Password"
        android:inputType="textPassword"
        android:minHeight="48dp" />
```

```
    <EditText
        android:id="@+id/ageInput"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter Age"
        android:inputType="number"
        android:minHeight="48dp" />

    <EditText
        android:id="@+id/addressInput"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter Address"
        android:minHeight="48dp" />

    <EditText
        android:id="@+id/mobileInput"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter Mobile No"
        android:inputType="phone"
        android:minHeight="48dp" />

    <Button
        android:id="@+id/registerButton"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Register"
        android:layout_marginTop="16dp"
        android:minHeight="48dp" />

    <TextView
        android:id="@+id/resultText"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:textSize="16sp"
        android:textStyle="bold"
        android:layout_marginTop="16dp" />
</LinearLayout>

-------------------MainActivity.java---------------------

package com.example.registrationform;

import android.os.Bundle;
import android.text.TextUtils;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.TextView;
import android.widget.Toast;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {
    private EditText nameInput, emailInput, passwordInput, ageInput, addressInput,
mobileInput;
    private Button registerButton;
    private TextView resultText;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
```

```
        nameInput = findViewById(R.id.nameInput);
        emailInput = findViewById(R.id.emailInput);
        passwordInput = findViewById(R.id.passwordInput);
        ageInput = findViewById(R.id.ageInput);
        addressInput = findViewById(R.id.addressInput);
        mobileInput = findViewById(R.id.mobileInput);
        registerButton = findViewById(R.id.registerButton);
        resultText = findViewById(R.id.resultText);

        registerButton.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                validateAndRegister();
            }
        });
    }

    private void validateAndRegister() {
        String name = nameInput.getText().toString().trim();
        String email = emailInput.getText().toString().trim();
        String password = passwordInput.getText().toString().trim();
        String age = ageInput.getText().toString().trim();
        String address = addressInput.getText().toString().trim();
        String mobile = mobileInput.getText().toString().trim();

        if (TextUtils.isEmpty(name)) {
            nameInput.setError("Name is required");
            return;
        }

        if (TextUtils.isEmpty(email) ||
!android.util.Patterns.EMAIL_ADDRESS.matcher(email).matches()) {
            emailInput.setError("Valid email is required");
            return;
        }

        if (TextUtils.isEmpty(password) || password.length() < 6) {
            passwordInput.setError("Password must be at least 6 characters long");
            return;
        }

        if (TextUtils.isEmpty(age) || !age.matches("\\d+") || Integer.parseInt(age) <= 0) {
            ageInput.setError("Valid age is required");
            return;
        }

        if (TextUtils.isEmpty(address)) {
            addressInput.setError("Address is required");
            return;
        }

        if (TextUtils.isEmpty(mobile) || !mobile.matches("\\d{10}")) {
            mobileInput.setError("Valid 10-digit mobile number is required");
            return;
        }

        resultText.setText("Registration Successful\nName: " + name + "\nEmail: " + email +
"\nAge: " + age + "\nAddress: " + address + "\nMobile: " + mobile);
        Toast.makeText(this, "Registration Successful", Toast.LENGTH_SHORT).show();
    }
}
```

Q.2] Write a Java Android Program to Demonstrate List View Activity with all operations Such as: Insert, Delete, Search

---------------activity_main.xml-----------------

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="16dp">

    <EditText
        android:id="@+id/editText"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter item"
        android:minHeight="48dp" />

    <Button
        android:id="@+id/addButton"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Add Item" />

    <Button
        android:id="@+id/deleteButton"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Delete Item" />

    <Button
        android:id="@+id/searchButton"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Search Item" />

    <ListView
        android:id="@+id/listView"
        android:layout_width="match_parent"
        android:layout_height="wrap_content" />
</LinearLayout>

-------------------MainActivity.java---------------------

package com.example.listviewactivity;
import android.os.Bundle;
import android.view.View;
import android.widget.AdapterView;
import android.widget.ArrayAdapter;
import android.widget.Button;
import android.widget.EditText;
import android.widget.ListView;
import android.widget.Toast;
import androidx.appcompat.app.AppCompatActivity;
import java.util.ArrayList;

public class MainActivity extends AppCompatActivity {

    private ListView listView;
    private EditText editText;
    private Button addButton, deleteButton, searchButton;
    private ArrayList<String> itemList;
    private ArrayAdapter<String> adapter;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
```

```
        listView = findViewById(R.id.listView);
        editText = findViewById(R.id.editText);
        addButton = findViewById(R.id.addButton);
        deleteButton = findViewById(R.id.deleteButton);
        searchButton = findViewById(R.id.searchButton);

        itemList = new ArrayList<>();
        adapter = new ArrayAdapter<>(this, android.R.layout.simple_list_item_1, itemList);
        listView.setAdapter(adapter);

        // Add item to the list
        addButton.setOnClickListener(v -> {
            String item = editText.getText().toString().trim();
            if (!item.isEmpty()) {
                itemList.add(item);
                adapter.notifyDataSetChanged();
                editText.setText("");
            } else {
                Toast.makeText(MainActivity.this, "Enter an item",
Toast.LENGTH_SHORT).show();
            }
        });

        // Delete selected item from the list
        listView.setOnItemClickListener((parent, view, position, id) -> {
            itemList.remove(position);
            adapter.notifyDataSetChanged();
        });

        deleteButton.setOnClickListener(v -> {
            String item = editText.getText().toString().trim();
            if (itemList.remove(item)) {
                adapter.notifyDataSetChanged();
                Toast.makeText(MainActivity.this, "Item deleted",
Toast.LENGTH_SHORT).show();
            } else {
                Toast.makeText(MainActivity.this, "Item not found",
Toast.LENGTH_SHORT).show();
            }
        });

        // Search item in the list
        searchButton.setOnClickListener(v -> {
            String item = editText.getText().toString().trim();
            if (itemList.contains(item)) {
                Toast.makeText(MainActivity.this, "Item found", Toast.LENGTH_SHORT).show();
            } else {
                Toast.makeText(MainActivity.this, "Item not found",
Toast.LENGTH_SHORT).show();
            }
        });
    }
}
```

Slip 14
Q.1] Construct an Android application to accept a number and calculate and display Factorial of a given number in TextView.

```
---------------------------------------activity_main.xml------------------------------------
---------------------------
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
```

```
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <EditText
        android:id="@+id/numberInput"
        android:layout_width="0dp"
        android:layout_height="wrap_content"
        android:hint="Enter a number"
        android:inputType="number"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent" />

    <Button
        android:id="@+id/calculateButton"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Calculate Factorial"
        app:layout_constraintTop_toBottomOf="@+id/numberInput"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent" />

    <TextView
        android:id="@+id/factorialResult"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Factorial will appear here"
        app:layout_constraintTop_toBottomOf="@+id/calculateButton"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintBottom_toBottomOf="parent" />

</androidx.constraintlayout.widget.ConstraintLayout>
```

---MainActivity.java---

```
package com.example.slip14;

import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.TextView;

import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        // Declare UI elements
        EditText numberInput = findViewById(R.id.numberInput);
        Button calculateButton = findViewById(R.id.calculateButton);
        TextView factorialResult = findViewById(R.id.factorialResult);

        // Set the button click listener
```

```
        calculateButton.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                String numberString = numberInput.getText().toString();

                if (!numberString.isEmpty()) {
                    try {
                        int number = Integer.parseInt(numberString);
                        if (number >= 0) {
                            // Calculate factorial
                            long result = factorial(number);
                            // Display result
                            factorialResult.setText("Factorial of " + number + " is: " +
result);
                        } else {
                            // Display error for negative numbers
                            factorialResult.setText("Please enter a valid positive number");
                        }
                    } catch (NumberFormatException e) {
                        // Handle non-numeric input
                        factorialResult.setText("Please enter a valid number");
                    }
                } else {
                    factorialResult.setText("Please enter a number");
                }
            }
        });
    }

    // Function to calculate factorial
    private long factorial(int n) {
        long result = 1;
        for (int i = 1; i <= n; i++) {
            result *= i;
        }
        return result;
    }
}
```

Q.2] Create an Android application, which show Login Form. After clicking LOGIN button display the â€œLogin Successfulâ€¦â€ message if username and password is same else display â€œInvalid Loginâ€ message in Toast Control

```
-----------------------------------------------activity_main.xml----------------------------
------------------------------
```

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <!-- Login Form Title (Text above the form) -->
    <TextView
        android:id="@+id/loginTitle"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Login Form"
        android:textSize="24sp"
        android:textColor="#000000"
        android:layout_marginTop="100dp"/>

    <!-- Username Input -->
```

```
    <EditText
        android:id="@+id/usernameInput"
        android:layout_width="0dp"
        android:layout_height="wrap_content"
        android:hint="Username"
        app:layout_constraintTop_toBottomOf="@+id/loginTitle"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        android:layout_marginTop="16dp"/>

    <!-- Password Input -->
    <EditText
        android:id="@+id/passwordInput"
        android:layout_width="0dp"
        android:layout_height="wrap_content"
        android:hint="Password"
        android:inputType="textPassword"
        app:layout_constraintTop_toBottomOf="@+id/usernameInput"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        android:layout_marginTop="16dp" />

    <!-- Login Button -->
    <Button
        android:id="@+id/loginButton"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="LOGIN"
        app:layout_constraintTop_toBottomOf="@+id/passwordInput"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        android:layout_marginTop="20dp" />

</androidx.constraintlayout.widget.ConstraintLayout>


--------------------------------------------------------------------------------------------
----------------

----------------------------------MainActivity.java-----------------------------------------
-----------------

package com.example.slip14q2;

import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.Toast;

import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    // Predefined username and password
    private static final String CORRECT_USERNAME = "admin";
    private static final String CORRECT_PASSWORD = "password123";

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main); // Ensure the layout file is correctly
referenced

        // Initialize UI components
```

```
        EditText usernameInput = findViewById(R.id.usernameInput);
        EditText passwordInput = findViewById(R.id.passwordInput);
        Button loginButton = findViewById(R.id.loginButton);

        // Set the login button click listener
        loginButton.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                // Get username and password input values
                String username = usernameInput.getText().toString();
                String password = passwordInput.getText().toString();

                // Check if the credentials are correct
                if (username.equals(CORRECT_USERNAME) && password.equals(CORRECT_PASSWORD))
{
                    // Login successful
                    Toast.makeText(MainActivity.this, "Login Successful...",
Toast.LENGTH_SHORT).show();
                } else {
                    // Invalid login
                    Toast.makeText(MainActivity.this, "Invalid Login",
Toast.LENGTH_SHORT).show();
                }
            }
        });
    }
}
```

Slip 15

Q1] Construct an Android application to accept two numbers in two EditText, with four buttons as ADD, SUB, DIV and MULT and display Result using Toast Control.

---------------------------------------activity_main.xml---

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="20dp">

    <EditText
        android:id="@+id/etNumber1"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter first number"
        android:inputType="numberDecimal"
        android:padding="10dp"/>

    <EditText
        android:id="@+id/etNumber2"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter second number"
        android:inputType="numberDecimal"
        android:padding="10dp"/>

    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:orientation="horizontal"
        android:gravity="center">

        <Button
            android:id="@+id/btnAdd"
```

```
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="ADD"
            android:layout_margin="5dp"/>

        <Button
            android:id="@+id/btnSub"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="SUB"
            android:layout_margin="5dp"/>

        <Button
            android:id="@+id/btnMul"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="MULT"
            android:layout_margin="5dp"/>

        <Button
            android:id="@+id/btnDiv"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="DIV"
            android:layout_margin="5dp"/>
    </LinearLayout>

</LinearLayout>
```


---MainActivity.java------------------------------

```java
package com.example.slip15q1;

import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.Toast;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    EditText etNumber1, etNumber2;
    Button btnAdd, btnSub, btnMul, btnDiv;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        etNumber1 = findViewById(R.id.etNumber1);
        etNumber2 = findViewById(R.id.etNumber2);
        btnAdd = findViewById(R.id.btnAdd);
        btnSub = findViewById(R.id.btnSub);
        btnMul = findViewById(R.id.btnMul);
        btnDiv = findViewById(R.id.btnDiv);

        // Event listeners for buttons
        btnAdd.setOnClickListener(view -> performOperation("+"));
        btnSub.setOnClickListener(view -> performOperation("-"));
        btnMul.setOnClickListener(view -> performOperation("*"));
```

```
        btnDiv.setOnClickListener(view -> performOperation("/"));
    }

    private void performOperation(String operator) {
        String num1 = etNumber1.getText().toString().trim();
        String num2 = etNumber2.getText().toString().trim();

        if (num1.isEmpty() || num2.isEmpty()) {
            Toast.makeText(this, "Please enter both numbers", Toast.LENGTH_SHORT).show();
            return;
        }

        double number1 = Double.parseDouble(num1);
        double number2 = Double.parseDouble(num2);
        double result = 0;

        switch (operator) {
            case "+":
                result = number1 + number2;
                break;
            case "-":
                result = number1 - number2;
                break;
            case "*":
                result = number1 * number2;
                break;
            case "/":
                if (number2 == 0) {
                    Toast.makeText(this, "Cannot divide by zero!",
Toast.LENGTH_SHORT).show();
                    return;
                }
                result = number1 / number2;
                break;
        }

        // Display result in a Toast message
        Toast.makeText(this, "Result: " + result, Toast.LENGTH_LONG).show();
    }
}
```

Q2] Construct a bank app to display different menu like withdraw, deposit etc.

--activity_main.xml--

```
<?xml version="1.0" encoding="utf-8"?>
    <LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
        android:layout_width="match_parent"
        android:layout_height="match_parent"
        android:orientation="vertical"
        android:padding="16dp">
    <TextView
        android:id="@+id/tvTitle"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Simple Bank Application"
        android:textStyle="bold"
        android:textSize="18sp"
        android:layout_gravity="center_horizontal"/>

    <EditText
        android:id="@+id/etAccountNumber"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Account Number"
```

```xml
        android:inputType="number"
        android:minHeight="48dp"
        android:padding="12dp"/>

    <EditText
        android:id="@+id/etAccountType"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Account Type"
        android:minHeight="48dp"
        android:padding="12dp"/>

    <EditText
        android:id="@+id/etBalance"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Balance"
        android:minHeight="48dp"
        android:padding="12dp"
        android:inputType="numberDecimal" />

    <RadioGroup
        android:id="@+id/radioGroup"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:orientation="horizontal">

        <RadioButton
            android:id="@+id/rbChecking"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Checking"/>

        <RadioButton
            android:id="@+id/rbSavings"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Savings"/>
    </RadioGroup>

    <Button
        android:id="@+id/btnCreate"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Create" />

    <Button
        android:id="@+id/btnDeposit"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Deposit" />

    <Button
        android:id="@+id/btnWithdraw"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Withdraw" />

    <Button
        android:id="@+id/btnBalance"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Balance" />
    </LinearLayout>
```

------------------------------------MainActivity.java--

```java
package com.example.slip15q1;

import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.RadioButton;
import android.widget.RadioGroup;
import android.widget.Toast;

import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    private EditText etAccountNumber, etAccountType, etBalance;
    private RadioGroup radioGroup;
    private Button btnCreate, btnDeposit, btnWithdraw, btnBalance;

    private double balance = 0.0;
    private String accountType = "";

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        etAccountNumber = findViewById(R.id.etAccountNumber);
        etAccountType = findViewById(R.id.etAccountType);
        etBalance = findViewById(R.id.etBalance);
        radioGroup = findViewById(R.id.radioGroup);

        btnCreate = findViewById(R.id.btnCreate);
        btnDeposit = findViewById(R.id.btnDeposit);
        btnWithdraw = findViewById(R.id.btnWithdraw);
        btnBalance = findViewById(R.id.btnBalance);

        btnCreate.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                createAccount();
            }
        });

        btnDeposit.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                depositMoney();
            }
        });

        btnWithdraw.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                withdrawMoney();
            }
        });
```

```
        btnBalance.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                checkBalance();
            }
        });

    }

    private void createAccount() {
        if (etAccountNumber.getText().toString().isEmpty() ||
etAccountType.getText().toString().isEmpty()) {
            Toast.makeText(this, "Please enter account details!",
Toast.LENGTH_SHORT).show();
            return;
        }

        int selectedId = radioGroup.getCheckedRadioButtonId();
        if (selectedId == -1) {
            Toast.makeText(this, "Please select account type!", Toast.LENGTH_SHORT).show();
            return;
        }

        RadioButton selectedRadioButton = findViewById(selectedId);
        accountType = selectedRadioButton.getText().toString();
        balance = 0.0;
        etBalance.setText(String.valueOf(balance));

        Toast.makeText(this, "Account Created Successfully!", Toast.LENGTH_SHORT).show();
    }

    private void depositMoney() {
        double amount = getAmountInput();
        if (amount <= 0) {
            return;
        }

        balance += amount;
        etBalance.setText(String.valueOf(balance));
        Toast.makeText(this, "Deposited: â,¹" + amount, Toast.LENGTH_SHORT).show();
    }

    private void withdrawMoney() {
        double amount = getAmountInput();
        if (amount <= 0) {
            return;
        }

        if (amount > balance) {
            Toast.makeText(this, "Insufficient balance!", Toast.LENGTH_SHORT).show();
        } else {
            balance -= amount;
            etBalance.setText(String.valueOf(balance));
            Toast.makeText(this, "Withdrawn: â,¹" + amount, Toast.LENGTH_SHORT).show();
        }
    }

    private void checkBalance() {
        Toast.makeText(this, "Current Balance: â,¹" + balance, Toast.LENGTH_SHORT).show();
    }

    private double getAmountInput() {
        String input = etBalance.getText().toString();
        if (input.isEmpty()) {
            Toast.makeText(this, "Enter a valid amount!", Toast.LENGTH_SHORT).show();
```

```
            return -1;
        }
        return Double.parseDouble(input);
    }
}
```

Slip 16
Q1] Create a Simple Android Application Which Send â€•Helloâ€– message from one activity to another with help of Button (Use Intent).

----------------activity_main.xml----------------------

```
<?xml version="1.0" encoding="utf-8"?>
<FrameLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <!-- Layout for the Main Activity: Button to send the message -->
    <LinearLayout
        android:id="@+id/mainLayout"
        android:layout_width="match_parent"
        android:layout_height="match_parent"
        android:orientation="vertical"
        android:gravity="center"
        android:padding="16dp">

        <Button
            android:id="@+id/sendButton"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Send Hello" />
    </LinearLayout>

    <!-- Layout for the Second Activity: TextView to display the message -->
    <LinearLayout
        android:id="@+id/secondLayout"
        android:layout_width="match_parent"
        android:layout_height="match_parent"
        android:orientation="vertical"
        android:gravity="center"
        android:padding="16dp"
        android:visibility="gone">

        <TextView
            android:id="@+id/messageTextView"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Message will appear here"
            android:textSize="18sp"/>
    </LinearLayout>
</FrameLayout>
```

------------MainActivity.java----------

```
package com.example.slip16_q1;

import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.TextView;
import android.widget.LinearLayout;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    private LinearLayout mainLayout, secondLayout;
    private Button sendButton;
    private TextView messageTextView;
```

```java
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        // Initialize the views
        mainLayout = findViewById(R.id.mainLayout);
        secondLayout = findViewById(R.id.secondLayout);
        sendButton = findViewById(R.id.sendButton);
        messageTextView = findViewById(R.id.messageTextView);

        // Set button click listener
        sendButton.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                // Hide main layout and show second layout
                mainLayout.setVisibility(View.GONE);
                secondLayout.setVisibility(View.VISIBLE);

                // Set the message in the TextView
                messageTextView.setText("Hello");
            }
        });
    }
}
```

Q2] Create an Android application, with two activity first activity will have an EditText and a Button where the user can enter player name and after clicking on button the entered name will be display in another Activity. Second activity has the BACK button to transition to first activity (Using Intent).

```xml
---activity_main.xml---------
<?xml version="1.0" encoding="utf-8"?>
<FrameLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <!-- Layout for the First Activity: Contains EditText and Button -->
    <LinearLayout
        android:id="@+id/firstActivityLayout"
        android:layout_width="match_parent"
        android:layout_height="match_parent"
        android:orientation="vertical"
        android:gravity="center"
        android:padding="16dp">

        <EditText
            android:id="@+id/editTextName"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:hint="Enter Player Name"
            android:inputType="textPersonName" />

        <Button
            android:id="@+id/buttonSend"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Send Name"
            android:layout_marginTop="20dp" />
    </LinearLayout>

    <!-- Layout for the Second Activity: Contains TextView and Back Button -->
    <LinearLayout
        android:id="@+id/secondActivityLayout"
        android:layout_width="match_parent"
```

```
        android:layout_height="match_parent"
        android:orientation="vertical"
        android:gravity="center"
        android:padding="16dp"
        android:visibility="gone">

        <TextView
            android:id="@+id/textViewName"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Player Name"
            android:textSize="18sp" />

        <Button
            android:id="@+id/buttonBack"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="BACK"
            android:layout_marginTop="20dp" />
    </LinearLayout>
</FrameLayout>
```

----------------------MainActivity.java-------------

```
package com.example.slip16_q2;

import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.LinearLayout;
import android.widget.TextView;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    // Declare views for both activities
    private LinearLayout firstActivityLayout, secondActivityLayout;
    private EditText editTextName;
    private Button buttonSend, buttonBack;
    private TextView textViewName;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        // Initialize the views
        firstActivityLayout = findViewById(R.id.firstActivityLayout);
        secondActivityLayout = findViewById(R.id.secondActivityLayout);
        editTextName = findViewById(R.id.editTextName);
        buttonSend = findViewById(R.id.buttonSend);
        buttonBack = findViewById(R.id.buttonBack);
        textViewName = findViewById(R.id.textViewName);

        // Button click listener to transition from First Activity to Second Activity
        buttonSend.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                // Get the entered player name
                String playerName = editTextName.getText().toString();

                // Set the player's name in the TextView of Second Activity layout
                textViewName.setText(playerName);
```

```
                // Hide the first activity layout and show the second activity layout
                firstActivityLayout.setVisibility(View.GONE);
                secondActivityLayout.setVisibility(View.VISIBLE);
            }
        });

        // Button click listener to go back to the First Activity
        buttonBack.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                // Show the first activity layout and hide the second activity layout
                firstActivityLayout.setVisibility(View.VISIBLE);
                secondActivityLayout.setVisibility(View.GONE);
            }
        });
    }
}
```

Slip 18

Q1] Create an Android Application that will change color of the screen and change the font size of text view using xml.

```
---activity_main.xml---------
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
            xmlns:app="http://schemas.android.com/apk/res-auto"
            xmlns:tools="http://schemas.android.com/tools"
            android:layout_width="match_parent"
            android:layout_height="match_parent"
            android:background="#FF0000"
    tools:ignore="ExtraText"> <!-- Set background color to yellow -->

                <!-- TextView to display some text -->
                <TextView
                    android:id="@+id/textView"
                    android:layout_width="wrap_content"
                    android:layout_height="wrap_content"
                    android:text="Hello, Android!"
                    android:textSize="40sp"
                    tools:ignore="MissingConstraints" />    <!-- Set the font size to 24sp -
->
                android:textColor="#FFFFFF"                 <!-- Set text color to white -->
                app:layout_constraintTop_toTopOf="parent"          <!-- Align top of
TextView to top of parent -->
                app:layout_constraintBottom_toBottomOf="parent"  <!-- Align bottom of
TextView to bottom of parent -->
                app:layout_constraintLeft_toLeftOf="parent" <!-- Align left side of TextView
to left of parent -->
                app:layout_constraintRight_toRightOf="parent" <!-- Align right side of
TextView to right of parent -->
                app:layout_constraintHorizontal_bias="0.5"  <!-- Horizontally center the
TextView -->
                app:layout_constraintVertical_bias="0.5"    <!-- Vertically center the
TextView -->
                android:padding="16dp"/>

        </androidx.constraintlayout.widget.ConstraintLayout>

----------------------MainActivity.java-------------

package com.example.slip18q1;

import androidx.appcompat.app.AppCompatActivity;
```

```
import android.os.Bundle;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
    }
}
```

Q2] Create table Project (id, name, dept, city). Create Application to perform the following operations. (using SQLite database)
i] Add at least 5 records.
ii] Display all the records.

```
--------- activity_main.xml ------------
<?xml version="1.0"  encoding="utf-8"?>
<LinearLayout
     xmlns:android="https://schemas.android.com/apk/res/android"
     xmlns:android="https://schemas.android.com/apk/res-auto"
     xmlns:tools="https://schemas.android.com/tools"
     android:layout_width="match_parent"
     android:layout_height="match_parent"
     android:orientation="vertical"
     tools:context=".MainActivity">

<EditText
        android:layout_width="300dp"
        android:layout_height="wrap_content"
        android:id="@+id/et1"
        android:hint="Enter Project Id"/>

<EditText
        android:layout_width="300dp"
        android:layout_height="wrap_content"
        android:id="@+id/et2"
        android:hint="Enter Projec Name"/>

<EditText
        android:layout_width="300dp"
        android:layout_height="wrap_content"
        android:id="@+id/et3"
        android:hint="Enter Department` "/>

<EditText
        android:layout_width="300dp"
        android:layout_height="wrap_content"
        android:id="@+id/et4"
        android:hint="Enter City"/>

<Button
        android:layout_width="300dp"
        android:layout_height="300dp"
        android:id="@+id/bt1"
        android:hint="Save"/>

<Button
        android:layout_width="300dp"
        android:layout_height="300dp"
        android:id="@+id/bt"
        android:hint="Show"/>

<ListView
        android:layout_width="match_parent"
```

```
        android:layout_height="match_parent"
        android:id="@+id/lv"/>

</LinearLayout>
```

--------- MainActivity.java ------------

```
package com.example.database;

import androidx.appcompat.app.AppCompatActivity;

import android.os.Bundle;
import android.view.View;
import android.widget.ArrayAdapter;
import android.widget.Button;
import android.widget.EditText;
import android.widget.ListView;
import android.widget.Toast;

import java.util.ArrayList;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        mydb db = new mydb(MainActivity.this);
        ListView l = (ListView) findViewById(R.id.lv);
        EditText epid =(EditText) findViewById(R.id.et1);
        EditText epname =(EditText) findViewById(R.id.et2);
        EditText epdept =(EditText) findViewById(R.id.et3);
        EditText epcity =(EditText) findViewById(R.id.et4);
        Button b = (Button) findViewById(R.id.bt1);
        Button b2 = (Button) findViewById(R.id.bt2);
        b.setOnClickListener(new View.OnClickListener()
        {
            @Override
            public void onClick(View v)
            {
                final String id = epid.getText().toString();
                final String name = epname.getText().toString();
                final String dept = epdept.getText().toString();
                final String city = epcity.getText().toString();
                long retValue = db.addDetails(id,name,dept,city);
                if(retValue >= 0)
                    Toast.makeText(getApplicationContext(),"Data Saved
Successfully",Toast.LENGTH_LONG).show();
                else
                    Toast.makeText(getApplicationContext(),"Unable to save
data",Toast.LENGTH_LONG).show();
            }
        });
        b2.setOnClickListener(new View.OnClickListener()
        {
            @Override
            public void onClick(View v)
            {
                String s="";
                ArrayList<String> c=db.getDetails(getApplicationContext());
                /*Iterator itr = c.iterator();
                while(itr.hasNext())
                {

Toast.makeText(getApplicationContext(),itr.next().toString(),Toast.LENGTH_LONG).show();
```

```
                */
                ArrayAdapter<String> aa = new ArrayAdapter<String>
(getApplicationContext(),android.R.layout.simple_list_item_1,c);
                l.setAdapter(aa);
            }
        });
    }
}



------------ mydb.java -------------
package com.example.database;

import android.content.ContentValues;
import android.content.Context;
import android.database.Cursor;
import android.database.sqlite.SQLiteDatabase;
import android.database.sqlite.SQLiteOpenHelper;

import java.util.ArrayList;
import java.util.Collections;

public class mydb<customer> extends SQLiteOpenHelper
{
    static String dbname="student.db";
    public static final int dbversion=1;
    public mydb(Context c)
    {
        super(c,dbname,null,dbversion);
    }
    public mydb(Context c, String name, SQLiteDatabase.CursorFactory factory,int version)
    {
        super(c,name,factory,version);
    }
    @Override
    public void onCreate(SQLiteDatabase dba)
    {
        dba.execSQL("create table project(pid integer PRIMARY KEY,pname text,pdept text,city
text)");
    }
    @Override
    public void onUpgrade(SQLiteDatabase db,int oidVersion,int newVersion)
    {

    }
    public ArrayList<String> getDetails(Context c11)
    {
        ArrayList<String> s1 = new ArrayList<String>(Collections.singleton(""));
        String sql = "select * from project";
        SQLiteDatabase db = this.getReadableDatabase();
        Cursor c = db.rawQuery(sql,null);
        if(c.getCount() > 0)
        {
            c.moveToFirst(); do
            {
                Integer id = c.getString(c.getColumnIndexOrThrow("pid"));
                String name = c.getString(c.getColumnIndexOrThrow("pname"));
                String dept = c.getString(c.getColumnIndexOrThrow("pdept"));
                String city = c.getString(c.getColumnIndexOrThrow("city"));
                String c1 ="Project Id : "+ id + "\n" + "Project Name : "+ name + "\n
"+"Department : " + dept+ " \n"+ "City : "+ city+"\n";
                s1.add(c1);
            }
            while(c.moveToNext());
```

```
        }
        return s1;
    }
    public long addDetails(Integer id,String name,String dept,String city)
    {
        ContentValues cv = new ContentValues();
        cv.put("pid",id);
        cv.put("pname",name);
        cv.put("pdept",dept);
        cv.put("city",city);
        SQLiteDatabase db = this.getWritableDatabase();
        long retValue = db.insert("project",null,cv);
        return retValue;
    }
}


----------- AndroidManifest.xml --------------
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools">

    <application
        android:allowBackup="true"
        android:dataExtractionRules="@xml/data_extraction_rules"
        android:fullBackupContent="@xml/backup_rules"
        android:icon="@mipmap/ic_launcher"
        android:label="@string/app_name"
        android:roundIcon="@mipmap/ic_launcher_round"
        android:supportsRtl="true"
        android:theme="@style/Theme.Database"
        tools:targetApi="31">
        <activity
            android:name=".MainActivity"
            android:exported="true">
            <intent-filter>
                <action android:name="android.intent.action.MAIN" />

                <category android:name="android.intent.category.LAUNCHER" />
            </intent-filter>
        </activity>
    </application>

</manifest>


Slip 19
Q1] Write an Android Program to Change the Image Displayed on the Screen.

---activity_main.xml---------

<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <TextView
        android:id="@+id/textView"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
```

```
        android:layout_marginStart="50dp"
        android:layout_marginTop="10dp"
        android:layout_marginEnd="50dp"
        android:layout_marginBottom="40dp"
        android:text="@string/change_image_demo"
        app:layout_constraintBottom_toTopOf="@+id/IView"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintVertical_bias="0.042" />

    <ImageView
        android:id="@+id/imageView"
        android:layout_width="300dp"
        android:layout_height="300dp"
        android:layout_marginStart="30dp"
        android:layout_marginTop="40dp"
        android:layout_marginEnd="30dp"
        android:layout_marginBottom="50dp"
        android:contentDescription="@string/todo"
        android:src="@drawable/rose1"
        app:layout_constraintBottom_toTopOf="@+id/changeImageButton"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toBottomOf="@+id/textView"
        tools:ignore="MissingConstraints" />

    <Button
        android:id="@+id/changeImageButton"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_below="@id/IView"
        android:layout_centerHorizontal="true"
        android:layout_marginStart="159dp"
        android:layout_marginTop="40dp"
        android:layout_marginEnd="111dp"
        android:layout_marginBottom="50dp"
        android:text="@string/change_image"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toBottomOf="@+id/IView"
        tools:ignore="MissingConstraints" />
</androidx.constraintlayout.widget.ConstraintLayout>

----------------------MainActivity.java-------------

package com.example.imagechange;

import android.annotation.SuppressLint;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.ImageView;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    private ImageView imageView;

    @SuppressLint("MissingInflatedId")
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
```

```
        setContentView(R.layout.activity_main);

        // Initialize views
        imageView = findViewById(R.id.imageView);
        Button changeImageButton = findViewById(R.id.changeImageButton);

        // Set OnClickListener to change the image when button is clicked
        changeImageButton.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                // Change image to a new one when button is clicked
                imageView.setImageResource(R.drawable.rose2); // Replace with your new image
resource
            }
        });
    }
}
```

Q2] Construct an Android Application to create two option menu as Find Factorial and Find Sum of Digits. Accept a number and calculate Factorial and Sum of Digits of a given number by clicking Menu.

```
---activity_main.xml---------
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <TextView
        android:id="@+id/textView"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginBottom="712dp"
        android:text="@string/factorial_and_sum_of_digit_program_app"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent" />

    <EditText
        android:id="@+id/etNumber"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginTop="119dp"
        android:layout_marginBottom="186dp"
        android:autofillHints=""
        android:hint="@string/enter_a_number"
        android:inputType="number"
        android:padding="10dp"
        app:layout_constraintBottom_toTopOf="@+id/tvResult"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toBottomOf="@+id/textView" />

    <TextView
        android:id="@+id/tvResult"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginTop="20dp"
        android:layout_marginBottom="319dp"
        android:paddingTop="20dp"
```

```
        android:text="@string/result_will_be_displayed_here"
        android:textSize="18sp"
        android:textStyle="bold"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toBottomOf="@+id/etNumber" />

</androidx.constraintlayout.widget.ConstraintLayout>


-------------MainActivity.java------

package com.example.factorialandsumapp;

import static com.example.factorialandsumapp.R.menu.main_menu;

import  com.example.factorialandsumapp.R;
import  com.example.factorialandsumapp.R;

import android.annotation.SuppressLint;
import android.os.Bundle;
import android.view.Menu;
import android.view.MenuInflater;
import android.view.MenuItem;
import android.widget.EditText;
import android.widget.TextView;
import android.widget.Toast;
import androidx.appcompat.app.AppCompatActivity;

import java.util.zip.Inflater;

public class MainActivity extends AppCompatActivity {

    EditText etNumber;
    TextView tvResult;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        etNumber = findViewById(R.id.etNumber);
        tvResult = findViewById(R.id.tvResult);
    }

    @Override
    public boolean onCreateOptionsMenu(Menu menu) {
        MenuInflater inflater = getMenuInflater(); // Get the menu inflater
        inflater.inflate(R.menu.main_menu, menu); // Correct way to inflate
        return true; // Return true to display th
    }

    private static final int MENU_FACTORIAL = 1;
    private static final int MENU_SUM_DIGITS = 2;
    public boolean onOptionsItemSelected (MenuItem item){
            String input = etNumber.getText().toString();

            if (input.isEmpty()) {
                Toast.makeText(this, "Please enter a number", Toast.LENGTH_SHORT).show();
                return true;
            }

            int number = Integer.parseInt(input);

            switch (item.getItemId()) {
```

```
                case MENU_FACTORIAL:
                    tvResult.setText("Factorial: " +
calculateFactorial(Integer.parseInt(etNumber.getText().toString())));
                    return true;

                case MENU_SUM_DIGITS:
                    tvResult.setText("Sum of Digits:" +
calculateSumOfDigits(Integer.parseInt(etNumber.getText().toString())));
                    return true;

                default:
                    return super.onOptionsItemSelected(item);
            }
        }

    private void setText(String s) {
    }

    // Function to calculate factorial
        private long calculateFactorial ( int num){
            if (num < 0) return -1; // Handle negative numbers

            long fact = 1;
            for (int i = 1; i <= num; i++) {
                fact *= i;
            }
            return fact;
        }

        // Function to calculate sum of digits
        private int calculateSumOfDigits ( int num){
            int sum = 0;
            while (num > 0) {
                sum += num % 10;
                num /= 10;
            }
            return sum;
        }
    }
```

Slip 21
Q.1] Write an Android Program to demonstrate Activity life Cycle.
---activity_main.xml---------

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity"/>
```

-------------MainActivity.java------

```
package com.example.myapplication12;

import androidx.appcompat.app.AppCompatActivity;

import android.os.Bundle;
import android.widget.Toast;

public class MainActivity extends AppCompatActivity {
```

```
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
    }
    protected void onStart()
    {
        Toast.makeText(this, "welcome", Toast.LENGTH_SHORT).show();
        super.onStart();
    }

    @Override
    protected void onResume()
    {
        Toast.makeText(this, "onResume() called", Toast.LENGTH_SHORT).show();
        super.onResume();
    }

    protected void onPause()
    {
        Toast.makeText(this, "onPause() called",Toast.LENGTH_SHORT).show();
        super.onPause();
    }
    protected void onStop()
    {
        Toast.makeText(this, "onStop() called",Toast.LENGTH_SHORT).show();
        super.onStop();
    }
    protected void onDestroy()
    {
        Toast.makeText(this, "onDestroy() called",Toast.LENGTH_SHORT).show();
        super.onDestroy();
    }
}
```

Q.2] Create an Android Application that writes data to the SD Card

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    android:padding="20dp">

    <EditText
        android:id="@+id/editTextData"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter text here"
        android:padding="10dp"/>

    <Button
        android:id="@+id/btnWrite"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="WRITE DATA"
        android:layout_marginTop="10dp"/>

    <Button
        android:id="@+id/btnClear"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="CLEAR"
```

```
        android:layout_marginTop="10dp"/>

    <TextView
        android:id="@+id/textStatus"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Status"
        android:textSize="16sp"
        android:textStyle="bold"
        android:gravity="center"
        android:layout_marginTop="20dp"/>

</LinearLayout>
```

```
package com.example.slip19q1;

import android.Manifest;
import android.content.pm.PackageManager;
import android.os.Bundle;
import android.os.Environment;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.TextView;
import android.widget.Toast;
import androidx.annotation.NonNull;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.app.ActivityCompat;
import java.io.File;
import java.io.FileOutputStream;
import java.io.IOException;

public class MainActivity extends AppCompatActivity {

    EditText editTextData;
    Button btnWrite, btnClear;
    TextView textStatus;

    private static final int REQUEST_PERMISSION = 1;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        editTextData = findViewById(R.id.editTextData);
        btnWrite = findViewById(R.id.btnWrite);
        btnClear = findViewById(R.id.btnClear);
        textStatus = findViewById(R.id.textStatus);

        // Request Storage Permission
        if (ActivityCompat.checkSelfPermission(this,
Manifest.permission.WRITE_EXTERNAL_STORAGE)
                != PackageManager.PERMISSION_GRANTED) {
            ActivityCompat.requestPermissions(this,
                    new String[]{Manifest.permission.WRITE_EXTERNAL_STORAGE},
REQUEST_PERMISSION);
        }

        btnWrite.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                writeDataToSDCard();
```

```
            }
        });

        btnClear.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                editTextData.setText("");
                textStatus.setText("Cleared");
            }
        });
    }

    private void writeDataToSDCard() {
        String text = editTextData.getText().toString();
        if (text.isEmpty()) {
            Toast.makeText(this, "Please enter text", Toast.LENGTH_SHORT).show();
            return;
        }

        if (isExternalStorageWritable()) {
            File sdCard = Environment.getExternalStorageDirectory();
            File dir = new File(sdCard.getAbsolutePath() + "/MyAppData");
            dir.mkdirs();
            File file = new File(dir, "data.txt");

            try (FileOutputStream fos = new FileOutputStream(file)) {
                fos.write(text.getBytes());
                textStatus.setText("Data Written in SDCARD");
                Toast.makeText(this, "Data saved to: " + file.getAbsolutePath(),
Toast.LENGTH_LONG).show();
            } catch (IOException e) {
                Toast.makeText(this, "Error writing file!", Toast.LENGTH_SHORT).show();
            }
        } else {
            Toast.makeText(this, "SD Card not available", Toast.LENGTH_SHORT).show();
        }
    }

    public boolean isExternalStorageWritable() {
        return Environment.getExternalStorageState().equals(Environment.MEDIA_MOUNTED);
    }

    // Handle Permission Result
    @Override
    public void onRequestPermissionsResult(int requestCode, @NonNull String[] permissions,
@NonNull int[] grantResults) {
        super.onRequestPermissionsResult(requestCode, permissions, grantResults);
        if (requestCode == REQUEST_PERMISSION) {
            if (grantResults.length > 0 && grantResults[0] ==
PackageManager.PERMISSION_GRANTED) {
                Toast.makeText(this, "Permission Granted", Toast.LENGTH_SHORT).show();
            } else {
                Toast.makeText(this, "Permission Denied", Toast.LENGTH_SHORT).show();
            }
        }
    }
}
```

Slip 22
Q.1] Write an Java Android Program to Change the Image on the Screen.
---activity_main.xml---------

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
```

```
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <TextView
        android:id="@+id/textView"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginStart="50dp"
        android:layout_marginTop="10dp"
        android:layout_marginEnd="50dp"
        android:layout_marginBottom="40dp"
        android:text="@string/change_image_demo"
        app:layout_constraintBottom_toTopOf="@+id/IView"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintVertical_bias="0.042" />

    <ImageView
        android:id="@+id/imageView"
        android:layout_width="300dp"
        android:layout_height="300dp"
        android:layout_marginStart="30dp"
        android:layout_marginTop="40dp"
        android:layout_marginEnd="30dp"
        android:layout_marginBottom="50dp"
        android:contentDescription="@string/todo"
        android:src="@drawable/rose1"
        app:layout_constraintBottom_toTopOf="@+id/changeImageButton"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toBottomOf="@+id/textView"
        tools:ignore="MissingConstraints" />

    <Button
        android:id="@+id/changeImageButton"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_below="@id/IView"
        android:layout_centerHorizontal="true"
        android:layout_marginStart="159dp"
        android:layout_marginTop="40dp"
        android:layout_marginEnd="111dp"
        android:layout_marginBottom="50dp"
        android:text="@string/change_image"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toBottomOf="@+id/IView"
        tools:ignore="MissingConstraints" />
</androidx.constraintlayout.widget.ConstraintLayout>
```

----------------------MainActivity.java-------------

```
package com.example.imagechange;

import android.annotation.SuppressLint;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
```

```java
import android.widget.ImageView;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    private ImageView imageView;

    @SuppressLint("MissingInflatedId")
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);

        setContentView(R.layout.activity_main);

        // Initialize views
        imageView = findViewById(R.id.imageView);
        Button changeImageButton = findViewById(R.id.changeImageButton);

        // Set OnClickListener to change the image when button is clicked
        changeImageButton.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                // Change image to a new one when button is clicked
                imageView.setImageResource(R.drawable.rose2); // Replace with your new image
resource
            }
        });
    }
}
```

Q.2] Perform following numeric operation according to user selection of radio button

---activity_main.xml---------

```xml
<?xml version="1.0" encoding="utf-8"?>

<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="20dp">

    <EditText
        android:id="@+id/etNumber"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter Number"
        android:inputType="number" />

    <RadioGroup
        android:id="@+id/radioGroup"
        android:layout_width="match_parent"
        android:layout_height="wrap_content">

        <RadioButton
            android:id="@+id/rbOddEven"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Odd or Even" />

        <RadioButton
            android:id="@+id/rbPosNeg"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Positive or Negative" />
```

```
        <RadioButton
            android:id="@+id/rbSquare"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Square" />
    </RadioGroup>

    <Button
        android:id="@+id/btnCheck"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Click" />

    <TextView
        android:id="@+id/tvResult"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Ans:"
        android:textSize="18sp"
        android:textStyle="bold"
        android:paddingTop="20dp"/>
</LinearLayout>
```

-------------MainActivity.java------

```
package com.example.slip22_2;

import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.RadioButton;
import android.widget.RadioGroup;
import android.widget.TextView;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {
    EditText etNumber;
    RadioGroup radioGroup;
    RadioButton rbOddEven, rbPosNeg, rbSquare;
    Button btnCheck;
    TextView tvResult;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        etNumber = findViewById(R.id.etNumber);
        radioGroup = findViewById(R.id.radioGroup);
        rbOddEven = findViewById(R.id.rbOddEven);
        rbPosNeg = findViewById(R.id.rbPosNeg);
        rbSquare = findViewById(R.id.rbSquare);
        btnCheck = findViewById(R.id.btnCheck);
        tvResult = findViewById(R.id.tvResult);

        btnCheck.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                String input = etNumber.getText().toString().trim();
                if (input.isEmpty()) {
                    tvResult.setText("Ans: Please enter a number");
                    return;
                }
```

```
                int number = Integer.parseInt(input);
                String result = "";

                if (rbOddEven.isChecked()) {
                    result = (number % 2 == 0) ? "Number is Even" : "Number is Odd";
                } else if (rbPosNeg.isChecked()) {
                    result = (number >= 0) ? "Number is Positive" : "Number is Negative";
                } else if (rbSquare.isChecked()) {
                    result = "Square: " + (number * number);
                } else {
                    result = "Please select an option";
                }

                tvResult.setText("Ans: " + result);
            }
        });
    }
}
```

Slip 23
Q. 1] Write a Java android program to demonstrate implicit intent

```
---activity_main.xml---------
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    android:padding="20dp">

    <Button
        android:id="@+id/btnOpenWebsite"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Open Website" />

</LinearLayout>

------------ AndroidManifest.xml ----------
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
    package="com.example.implicitintentdemo">

    <application
        android:allowBackup="true"
        android:theme="@style/Theme.ImplicitIntentDemo"
        android:usesCleartextTraffic="true">

        <activity android:name=".MainActivity">
            <intent-filter>
                <action android:name="android.intent.action.MAIN" />
                <category android:name="android.intent.category.LAUNCHER" />
            </intent-filter>
        </activity>

    </application>

</manifest>



-------------MainActivity.java------
package com.example.implicitintentdemo;
```

```
import android.content.Intent;
import android.net.Uri;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        Button btnOpenWebsite = findViewById(R.id.btnOpenWebsite);

        btnOpenWebsite.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                // Create an implicit intent
                Intent intent = new Intent(Intent.ACTION_VIEW);
                intent.setData(Uri.parse("https://www.google.com"));

                // Check if an activity exists to handle this intent
                if (intent.resolveActivity(getPackageManager()) != null) {
                    startActivity(intent);
                }
            }
   Â Â Â Â Â });
Â Â Â Â }
}
```

Q.2] Create an Android application which will ask the user to input his / her name. A message should display the two items concatenated in a label. Change the format of the label using radio buttons and check boxes for selection. The user can make the label text bold, underlined or italic as well as change its color. Also include buttons to display the message in the label, clear the text boxes as well as label. Finally exit.

---------------activity_main.xml-----------------

```
<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:padding="16dp">

    <!-- EditText for entering name -->
    <EditText
        android:id="@+id/editTextName"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter your name"
        android:layout_marginTop="50dp"
        android:padding="10dp"
        android:textSize="16sp"/>

    <!-- Button to display message -->
    <Button
        android:id="@+id/buttonDisplay"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Display Message"
        android:layout_below="@id/editTextName"
        android:layout_centerHorizontal="true"
        android:layout_marginTop="20dp"/>
```

```xml
<!-- Button to clear the text and label -->
<Button
    android:id="@+id/buttonClear"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Clear"
    android:layout_below="@id/buttonDisplay"
    android:layout_centerHorizontal="true"
    android:layout_marginTop="20dp"/>

<!-- Button to exit the app -->
<Button
    android:id="@+id/buttonExit"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Exit"
    android:layout_below="@id/buttonClear"
    android:layout_centerHorizontal="true"
    android:layout_marginTop="20dp"/>

<!-- TextView for displaying the message -->
<TextView
    android:id="@+id/textViewMessage"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Hello!"
    android:textSize="18sp"
    android:layout_below="@id/buttonExit"
    android:layout_centerHorizontal="true"
    android:layout_marginTop="20dp"/>

<!-- RadioGroup for text formatting -->
<RadioGroup
    android:id="@+id/radioGroupFormat"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_below="@id/textViewMessage"
    android:layout_marginTop="20dp"
    android:orientation="horizontal">

    <!-- Bold RadioButton -->
    <RadioButton
        android:id="@+id/radioBold"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Bold"/>

    <!-- Italic RadioButton -->
    <RadioButton
        android:id="@+id/radioItalic"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Italic"/>

    <!-- Underline RadioButton -->
    <RadioButton
        android:id="@+id/radioUnderline"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Underline"/>
</RadioGroup>

<!-- CheckBox for changing text color -->
<CheckBox
```

```
        android:id="@+id/checkBoxRed"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Red"
        android:layout_below="@id/radioGroupFormat"
        android:layout_marginTop="20dp"
        android:layout_alignParentLeft="true"/>

    <CheckBox
        android:id="@+id/checkBoxBlue"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Blue"
        android:layout_below="@id/radioGroupFormat"
        android:layout_marginTop="20dp"
        android:layout_alignParentRight="true"/>
</RelativeLayout>
```

-------------------MainActivity.java---------------------

```
package com.example.slip12_q2;

import android.graphics.Color;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.CheckBox;
import android.widget.EditText;
import android.widget.RadioButton;
import android.widget.RadioGroup;
import android.widget.TextView;
import android.widget.Toast;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    private EditText editTextName;
    private Button buttonDisplay, buttonClear, buttonExit;
    private TextView textViewMessage;
    private RadioGroup radioGroupFormat;
    private RadioButton radioBold, radioItalic, radioUnderline;
    private CheckBox checkBoxRed, checkBoxBlue;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        // Initialize the views
        editTextName = findViewById(R.id.editTextName);
        buttonDisplay = findViewById(R.id.buttonDisplay);
        buttonClear = findViewById(R.id.buttonClear);
        buttonExit = findViewById(R.id.buttonExit);
        textViewMessage = findViewById(R.id.textViewMessage);
        radioGroupFormat = findViewById(R.id.radioGroupFormat);
        radioBold = findViewById(R.id.radioBold);
        radioItalic = findViewById(R.id.radioItalic);
        radioUnderline = findViewById(R.id.radioUnderline);
        checkBoxRed = findViewById(R.id.checkBoxRed);
        checkBoxBlue = findViewById(R.id.checkBoxBlue);

        // Set the Display button's click listener
        buttonDisplay.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
```

```
                String name = editTextName.getText().toString().trim();
                if (name.isEmpty()) {
                    Toast.makeText(MainActivity.this, "Please enter your name",
Toast.LENGTH_SHORT).show();
                    return;
                }

                // Create the message
                String message = "Hello, " + name + "!";

                // Apply the selected text formatting
                if (radioBold.isChecked()) {
                    textViewMessage.setTypeface(null, android.graphics.Typeface.BOLD);
                } else if (radioItalic.isChecked()) {
                    textViewMessage.setTypeface(null, android.graphics.Typeface.ITALIC);
                } else if (radioUnderline.isChecked()) {
                    textViewMessage.setPaintFlags(textViewMessage.getPaintFlags() |
android.graphics.Paint.UNDERLINE_TEXT_FLAG);
                } else {
                    textViewMessage.setTypeface(null, android.graphics.Typeface.NORMAL);
                    textViewMessage.setPaintFlags(textViewMessage.getPaintFlags() &
(~android.graphics.Paint.UNDERLINE_TEXT_FLAG));
                }

                // Apply color formatting based on checkbox selection
                if (checkBoxRed.isChecked()) {
                    textViewMessage.setTextColor(Color.RED);
                } else if (checkBoxBlue.isChecked()) {
                    textViewMessage.setTextColor(Color.BLUE);
                } else {
                    textViewMessage.setTextColor(Color.BLACK);  // Default color
                }

                // Set the message in the TextView
                textViewMessage.setText(message);
            }
        });

        // Set the Clear button's click listener
        buttonClear.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                // Clear input fields and reset message
                editTextName.setText("");
                textViewMessage.setText("Hello!");
                radioGroupFormat.clearCheck();
                checkBoxRed.setChecked(false);
                checkBoxBlue.setChecked(false);
            }
        });

        // Set the Exit button's click listener
        buttonExit.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                // Exit the application
                finish();
            }
        });
    }
}
```

Slip 24

Q.1] Write an application to accept a string from the user. With two buttons to display the

string in Uppercase and Lowercase using the toast message.
------activity_main.xml----------
```
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    android:padding="20dp">

    <EditText
        android:id="@+id/editText"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter text"
        android:padding="10dp"/>

    <Button
        android:id="@+id/btnLower"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="LowerCase"
        android:layout_marginTop="10dp"/>

    <Button
        android:id="@+id/btnUpper"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="UpperCase"
        android:layout_marginTop="10dp"/>

</LinearLayout>
```

---------MainActivity.java-----------
```
package com.example.slip24_q1;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.Toast;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    private EditText editText;
    private Button btnLower, btnUpper;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        editText = findViewById(R.id.editText);
        btnLower = findViewById(R.id.btnLower);
        btnUpper = findViewById(R.id.btnUpper);

        btnLower.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                String input = editText.getText().toString();
                if (!input.isEmpty()) {
                    Toast.makeText(MainActivity.this, input.toLowerCase(),
Toast.LENGTH_SHORT).show();
                } else {
                    Toast.makeText(MainActivity.this, "Please enter text",
Toast.LENGTH_SHORT).show();
```

```
                }
            }
        });

        btnUpper.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                String input = editText.getText().toString();
                if (!input.isEmpty()) {
                    Toast.makeText(MainActivity.this, input.toUpperCase(),
Toast.LENGTH_SHORT).show();
                } else {
                    Toast.makeText(MainActivity.this, "Please enter text",
Toast.LENGTH_SHORT).show();
                }
            }
        });
    }
}
```

Q.2] Create table Car (id, name, type, color). Create Java Android Application for performing the following operation on the table. (Using SQLite database)
i)Insert 5 New Car Details.
ii)Show All the Car Details

```
--------- activity_main.xml ------------
<?xml version="1.0"  encoding="utf-8"?>
<LinearLayout
     xmlns:android="https://schemas.android.com/apk/res/android"
     xmlns:android="https://schemas.android.com/apk/res-auto"
     xmlns:tools="https://schemas.android.com/tools"
     android:layout_width="match_parent"
     android:layout_height="match_parent"
     android:orientation="vertical"
     tools:context=".MainActivity">

<EditText
        android:layout_width="300dp"
        android:layout_height="wrap_content"
        android:id="@+id/et1"
        android:hint="Enter Id"/>

<EditText
        android:layout_width="300dp"
        android:layout_height="wrap_content"
        android:id="@+id/et2"
        android:hint="Enter Name"/>

<EditText
        android:layout_width="300dp"
        android:layout_height="wrap_content"
        android:id="@+id/et3"
        android:hint="Enter Type"/>

<EditText
        android:layout_width="300dp"
        android:layout_height="wrap_content"
        android:id="@+id/et4"
        android:hint="Enter Color"/>

<Button
        android:layout_width="300dp"
        android:layout_height="300dp"
        android:id="@+id/bt1"
        android:hint="Save"/>
```

```
<Button
        android:layout_width="300dp"
        android:layout_height="300dp"
        android:id="@+id/bt"
        android:hint="Show"/>

<ListView
        android:layout_width="match_parent"
        android:layout_height="match_parent"
        android:id="@+id/lv"/>

</LinearLayout>


--------- MainActivity.java ------------
package com.example.database;

import androidx.appcompat.app.AppCompatActivity;

import android.os.Bundle;
import android.view.View;
import android.widget.ArrayAdapter;
import android.widget.Button;
import android.widget.EditText;
import android.widget.ListView;
import android.widget.Toast;

import java.util.ArrayList;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        mydb db = new mydb(MainActivity.this);
        ListView l = (ListView) findViewById(R.id.lv);
        EditText ecid =(EditText) findViewById(R.id.et1);
        EditText ecname =(EditText) findViewById(R.id.et2);
        EditText ectype =(EditText) findViewById(R.id.et3);
        EditText eccolor =(EditText) findViewById(R.id.et4);
        Button b = (Button) findViewById(R.id.bt1);
        Button b2 = (Button) findViewById(R.id.bt2);
        b.setOnClickListener(new View.OnClickListener()
        {
            @Override
            public void onClick(View v)
            {
                final String id = ecid.getText().toString();
                final String name = ecname.getText().toString();
                final String type = ectype.getText().toString();
                final String color = eccolor.getText().toString();
                long retValue = db.addDetails(rno,name,add,perc);
                if(retValue >= 0)
                    Toast.makeText(getApplicationContext(),"Data Saved
Successfully",Toast.LENGTH_LONG).show();
                else
                    Toast.makeText(getApplicationContext(),"Unable to save
data",Toast.LENGTH_LONG).show();
            }
        });
        b2.setOnClickListener(new View.OnClickListener()
        {
            @Override
```

```
            public void onClick(View v)
            {
                String s="";
                ArrayList<String> c=db.getDetails(getApplicationContext());
                /*Iterator itr = c.iterator();
                while(itr.hasNext())
                {

Toast.makeText(getApplicationContext(),itr.next().toString(),Toast.LENGTH_LONG).show();
                 */
                ArrayAdapter<String> aa = new ArrayAdapter<String>
(getApplicationContext(),android.R.layout.simple_list_item_1,c);
                l.setAdapter(aa);
            }
        });
    }
}



------------ mydb.java -------------
package com.example.database;

import android.content.ContentValues;
import android.content.Context;
import android.database.Cursor;
import android.database.sqlite.SQLiteDatabase;
import android.database.sqlite.SQLiteOpenHelper;

import java.util.ArrayList;
import java.util.Collections;

public class mydb<customer> extends SQLiteOpenHelper
{
    static String dbname="car.db";
    public static final int dbversion=1;
    public mydb(Context c)
    {
        super(c,dbname,null,dbversion);
    }
    public mydb(Context c, String name, SQLiteDatabase.CursorFactory factory,int version)
    {
        super(c,name,factory,version);
    }
    @Override
    public void onCreate(SQLiteDatabase dba)
    {
        dba.execSQL("create table car(cid integer PRIMARY KEY,cname text,ctype text,ccolor
text)");
    }
    @Override
    public void onUpgrade(SQLiteDatabase db,int oidVersion,int newVersion)
    {

    }
    public ArrayList<String> getDetails(Context c11)
    {
        ArrayList<String> s1 = new ArrayList<String>(Collections.singleton(""));
        String sql = "select * from car";
        SQLiteDatabase db = this.getReadableDatabase();
        Cursor c = db.rawQuery(sql,null);
        if(c.getCount() > 0)
        {
            c.moveToFirst(); do
            {
```

```
                String id = c.getString(c.getColumnIndexOrThrow("cid"));
                String name = c.getString(c.getColumnIndexOrThrow("cname"));
                String type = c.getString(c.getColumnIndexOrThrow("ctype"));
                String color = c.getString(c.getColumnIndexOrThrow("ccolor"));
                String c1 ="Car Id : "+ id + "\n" "Car Name : "+ name + "\n "+"Car Type : "
+ type+ " \n"+ "Car Color : "+ color+"\n";
                s1.add(c1);
            }
            while(c.moveToNext());
        }
        return s1;
    }
    public long addDetails(Integer id,String name,String type,String color)
    {
        ContentValues cv = new ContentValues();
        cv.put("cid",id);
        cv.put("cname",name);
        cv.put("ctype",type);
        cv.put("ccolor",color);
        SQLiteDatabase db = this.getWritableDatabase();
        long retValue = db.insert("car",null,cv);
        return retValue;
    }
}


----------- AndroidManifest.xml --------------
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools">

    <application
        android:allowBackup="true"
        android:dataExtractionRules="@xml/data_extraction_rules"
        android:fullBackupContent="@xml/backup_rules"
        android:icon="@mipmap/ic_launcher"
        android:label="@string/app_name"
        android:roundIcon="@mipmap/ic_launcher_round"
        android:supportsRtl="true"
        android:theme="@style/Theme.Database"
        tools:targetApi="31">
        <activity
            android:name=".MainActivity"
            android:exported="true">
            <intent-filter>
                <action android:name="android.intent.action.MAIN" />

                <category android:name="android.intent.category.LAUNCHER" />
            </intent-filter>
        </activity>
    </application>

</manifest>
```

Slip 25

Q.1] Create an android application for SMS activity.

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="20dp"
    android:gravity="center">

    <EditText
```

```
        android:id="@+id/etPhoneNumber"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter phone number"
        android:inputType="phone"
        android:padding="10dp"/>

    <EditText
        android:id="@+id/etMessage"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter message"
        android:padding="10dp"
        android:minLines="3"/>

    <Button
        android:id="@+id/btnSendSMS"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="SEND SMS"
        android:layout_marginTop="10dp"/>

</LinearLayout>
```

```
package com.example.slip25q1;


import android.Manifest;
import android.content.pm.PackageManager;
import android.os.Bundle;
import android.telephony.SmsManager;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.Toast;
import androidx.annotation.NonNull;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.app.ActivityCompat;
import androidx.core.content.ContextCompat;

public class MainActivity extends AppCompatActivity {

    EditText etPhoneNumber, etMessage;
    Button btnSendSMS;
    private static final int REQUEST_SMS_PERMISSION = 1;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        etPhoneNumber = findViewById(R.id.etPhoneNumber);
        etMessage = findViewById(R.id.etMessage);
        btnSendSMS = findViewById(R.id.btnSendSMS);

        btnSendSMS.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                if (ContextCompat.checkSelfPermission(MainActivity.this,
Manifest.permission.SEND_SMS)
                        != PackageManager.PERMISSION_GRANTED) {
                    ActivityCompat.requestPermissions(MainActivity.this,
                            new String[]{Manifest.permission.SEND_SMS},
```

```
REQUEST_SMS_PERMISSION);
                } else {
                    sendSMS();
                }
            }
        });
    }

    private void sendSMS() {
        String phoneNumber = etPhoneNumber.getText().toString();
        String message = etMessage.getText().toString();

        if (phoneNumber.isEmpty() || message.isEmpty()) {
            Toast.makeText(this, "Please enter phone number and message",
Toast.LENGTH_SHORT).show();
            return;
        }

        try {
            SmsManager smsManager = SmsManager.getDefault();
            smsManager.sendTextMessage(phoneNumber, null, message, null, null);
            Toast.makeText(this, "SMS Sent Successfully!", Toast.LENGTH_SHORT).show();
        } catch (Exception e) {
            Toast.makeText(this, "Failed to send SMS!", Toast.LENGTH_SHORT).show();
            e.printStackTrace();
        }
    }

    // Handle Permission Result
    @Override
    public void onRequestPermissionsResult(int requestCode, @NonNull String[] permissions,
@NonNull int[] grantResults) {
        super.onRequestPermissionsResult(requestCode, permissions, grantResults);
        if (requestCode == REQUEST_SMS_PERMISSION) {
            if (grantResults.length > 0 && grantResults[0] ==
PackageManager.PERMISSION_GRANTED) {
                sendSMS();
            } else {
                Toast.makeText(this, "Permission Denied!", Toast.LENGTH_SHORT).show();
            }
        }
    }
}
```

Q.2] Create an Android application, which show Login Form in table layout. After clicking LOGIN button display the â€œLogin Successfulâ€¦â€ message if username and password is same else display â€œInvalid Loginâ€ message in Toast Control.

---activity_main.xml---

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <!-- Login Form Title (Text above the form) -->
    <TextView
        android:id="@+id/loginTitle"
        android:layout_width="wrap_content"
```

```
        android:layout_height="wrap_content"
        android:text="Login Form"
        android:textSize="24sp"
        android:textColor="#000000"
        android:layout_marginTop="100dp"/>

    <!-- Username Input -->
    <EditText
        android:id="@+id/usernameInput"
        android:layout_width="0dp"
        android:layout_height="wrap_content"
        android:hint="Username"
        app:layout_constraintTop_toBottomOf="@+id/loginTitle"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        android:layout_marginTop="16dp"/>

    <!-- Password Input -->
    <EditText
        android:id="@+id/passwordInput"
        android:layout_width="0dp"
        android:layout_height="wrap_content"
        android:hint="Password"
        android:inputType="textPassword"
        app:layout_constraintTop_toBottomOf="@+id/usernameInput"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        android:layout_marginTop="16dp" />

    <!-- Login Button -->
    <Button
        android:id="@+id/loginButton"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="LOGIN"
        app:layout_constraintTop_toBottomOf="@+id/passwordInput"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        android:layout_marginTop="20dp" />

</androidx.constraintlayout.widget.ConstraintLayout>


--------------------------------------------------------------------------------------------
----------------

----------------------------------MainActivity.java-----------------------------------------
-----------------

package com.example.slip14q2;

import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.Toast;

import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    // Predefined username and password
    private static final String CORRECT_USERNAME = "admin";
    private static final String CORRECT_PASSWORD = "password123";
```

```
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main); // Ensure the layout file is correctly
referenced

        // Initialize UI components
        EditText usernameInput = findViewById(R.id.usernameInput);
        EditText passwordInput = findViewById(R.id.passwordInput);
        Button loginButton = findViewById(R.id.loginButton);

        // Set the login button click listener
        loginButton.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                // Get username and password input values
                String username = usernameInput.getText().toString();
                String password = passwordInput.getText().toString();

                // Check if the credentials are correct
                if (username.equals(CORRECT_USERNAME) && password.equals(CORRECT_PASSWORD))
{
                    // Login successful
                    Toast.makeText(MainActivity.this, "Login Successful...",
Toast.LENGTH_SHORT).show();
                } else {
                    // Invalid login
                    Toast.makeText(MainActivity.this, "Invalid Login",
Toast.LENGTH_SHORT).show();
                }
            }
        });
    }
}
```